ADDING POINTS AND LINES TO GENERALIZED POLYGONS: THE CONSTRUCTIONS WITHOUT COORDINATES

DEGREE–DIAMETER LAB

Abstract. Written for Gabriela Araujo-Pardo and Ignacio López. Each construction of the lab is described here only in terms of incidence: points, lines, the lines through a point, the points of a line, flags, anti-flags, collinearity, distances in the incidence graph, and configurations (ovoids, spreads, m-ovoids, whiskers, partial ovoids, local dualities) defined by their combinatorial properties. For each construction we say what is chosen in the starting geometry, which incidences are deleted or moved, which new points and lines are added and exactly what they are incident with, which pairs of elements stop being collinear or concurrent and how the new elements repair this, and why the diameter and the degrees are as claimed. Section 1 treats the diameter-3 constructions in a projective plane: the three hats of Araujo-Pardo and López and our four hats, side by side and in the same words. Sections 2–4 treat diameters 4, 6 and 7 in generalized quadrangles and hexagons, and Section 5 treats diameter 5: copies of a quadrangle joined by bijections, freed incidences and hubs, with an exact combinatorial criterion for which incidences can be freed. Section 6 treats Gómez's diameter-8 construction on copies of a generalized hexagon: it says what is his, why as defined its diameter is at least 9, and what is known about repairing it with a separate labelling of each copy (it works for $q = 2, 3$, fails for geometric labellings when $q \geq 7$, and is open otherwise). Section 7 treats our values at diameters 9 to 14, which do not come from a generalized polygon: they are Haar structures, with a point and a line for each element of a group and incidences given by a set S of Δ group elements. It says why two breadth-first searches decide the diameter, how S was found (closed under an automorphism of order 3, or random in a metacyclic group), why the orders found are a fraction about $1/\ln N$ of the Moore bound, and how each value was checked. The sections can be read independently.

Sources. Everything here is taken from the lab's notes: `research/theorems/POLISHED.md` (coordinates and complete proofs for diameters 3, 4 and 6), `research/theorems/D7_HEXAGON.md` (the same for diameter 7), `research/theorems/D5_QUADRANGLE.md` (diameter 5; the corrected version on branch `claude/d5-relax-searches-osr72a`, read with its referee report `research/referee/d5-quadrangle.md` on `claude/referee-d5-quadrangle-621u7o`), `research/theorems/OMEGA_NOTE.md` (diameter 8, branch `claude/omega-d8-note-xrbqux`; it replaces the note `papers/Ozden_OmegaH_D8_note.pdf`), `research/geometry/geometry.tex`, `research/plus2/coords/coords.tex` and the joint draft `research/paper/joint.tex`. No new result is claimed. Every combinatorial statement in Section 1 was checked by a program that builds both structures from the rules below on a plane given only by its incidence relation (`research/synthetic/checks/check_d3.py`: $\mathrm{PG}(2, q)$ for $q = 3, 4, 5, 7, 8, 9$, the two non-Desarguesian nearfield planes of order 9, and agreement with the coordinate version for $q \leq 9$). The other sections were checked in the same way on small quadrangles, hexagons and groups (Appendix A). Section 7 lists its own sources at its start.

0. Conventions

An *incidence structure* consists of points, lines and a symmetric incidence relation I between points and lines. Points and lines are *elements*. A *flag* is an incident point–line pair, an *anti-flag* a non-incident one. The *degree* of an element is the number of elements incident with it. Two distinct points are *collinear* if some line contains both, and two distinct lines are *concurrent* if some point

Date: 1 November 2026.

lies on both. We do *not* assume that two points lie on at most one line: after points and lines are added, two points may have several common lines, and two lines may have the same points.

The *incidence graph* has the elements as vertices and the flags as edges; it is bipartite. Distances $d(x, y)$ and the diameter are taken in it. So two points are at distance 2 when they are collinear, a point and a line are at distance 3 when they form an anti-flag and some point of the line is collinear with the point, and so on.

A *generalized n-gon* is an incidence structure whose incidence graph has diameter n and girth $2n$; elements at distance n are *opposite*. It has *order* (s, t) if every line has $s + 1$ points and every point is on $t + 1$ lines. Projective planes are the generalized 3-gons, generalized quadrangles the 4-gons, generalized hexagons the 6-gons. Write $n_b(\Delta, D)$ for the largest number of elements of an incidence structure with all degrees at most Δ and diameter at most D (the bipartite degree–diameter problem). A generalized n-gon of order (q, q) attains the counting (Moore) bound for $n_b(q + 1, n)$. When $q + 1$ is not a prime power, no generalized polygon of order $(q + 1, q + 1)$ is known, and every construction in Sections 1–6 starts from a generalized polygon of order (q, q) (or $(q - 1, q + 1)$) and adds points and lines, or joins several copies by new incidences, so that the maximum degree grows while the diameter stays the same or grows by one (by two in Section 6). Section 7 starts from a group instead. Table 1 is a summary.

Distances can be read off from collinearity alone:

Lemma 0.1 (Diameter in incidence terms)**.** *Let $\mathcal{S}$ be an incidence structure in which every element is incident with at least one element.*

(a) $\operatorname{diam}\mathcal{S} \le 3$ *if and only if any two points are collinear and any two lines are concurrent.*
(b) $\operatorname{diam}\mathcal{S} \le 4$ *if and only if for every anti-flag (p, L) some point of L is collinear with p.*
(c) $\operatorname{diam}\mathcal{S} \le 6$ *if and only if for every anti-flag (p, L) some point of L is at distance at most 4 from p.*
(d) $\operatorname{diam}\mathcal{S} \le 7$ *if and only if any two points, and any two lines, are at distance at most 6.*

Proof. For an even bound D: two points $p \ne p'$ are at distance at most D as soon as p is within $D-1$ of some line L through p' (if $p \mathrel{\mathrm{I}} L$ the distance is 2); two lines likewise. For an odd bound D: an anti-flag (p, L) is within distance D as soon as p is within $D - 1$ of some point of L. Distances between a point and a line are odd and distances between two points or two lines are even, which gives the converses. □

1. Diameter 3: hats in a projective plane

Let Π be a projective plane of order q: any two points are on exactly one line, any two lines meet in exactly one point, and every element has degree $q + 1$. For points $u \ne v$ we write uv for the line joining them. Both constructions in this section keep all points and lines of Π, delete or move a few flags near a chosen anti-flag, and add a few points and lines. By Lemma 0.1(a) the diameter stays 3 exactly when any two points remain collinear and any two lines remain concurrent; this is what has to be checked.

1.1. The frame.

Fix an anti-flag (A, ℓ). Every line through A meets ℓ in one point, and every point of ℓ is on one line through A. So the $q + 1$ flags (X, AX), X a point of ℓ, are the only incidences between the points of ℓ and the lines through A, and they match the points of ℓ with the lines through A. Both constructions change only some of these $q + 1$ flags.

Choose distinct *marked* points X_s on ℓ, indexed by a label set S, and put

$$\lambda_s = AX_s \qquad (s \in S).$$

One label $a \in S$ is distinguished, and $T = S \setminus \{a\}$. We write

$$\ell^* = \{\text{points of } \ell\} \setminus \{X_t : t \in T\}, \qquad \Lambda^* = \{\text{lines through } A\} \setminus \{\lambda_t : t \in T\}.$$

§	start	chosen configuration	changed incidences	added elements
1	projective plane of order q; $D = 3$, $\Delta = q + 2$	an anti-flag (A, ℓ) and 3 or 4 points of ℓ	1 flag deleted (three hats); 4 deleted and 3 added (four hats)	$4 + 4$ or $5 + 5$: hat points and hat lines, Q and M
2	generalized quadrangle; $D = 4$	a weighted m-ovoid and a weighted m'-cover (two elliptic quadrics in AS(q); a 2-ovoid and a spread of $W(q)$; …)	none	clones of points and shadow lines
3	generalized hexagon of order (q, q); $D = 6$	a spread Σ, a line $v \in \Sigma$, an exact whisker set	none; every element of three nested sets is replaced by several versions	$(m-1)\sum_k m^k \lvert\mathcal{A}_k\rvert$ versions
4	$2r + 1$ copies of a generalized hexagon; $D = 7$	bijections from the points of one copy to the lines of another; freed sets (segment sets, local dualities)	the incidences between copies at the freed points are deleted	hub points and hub lines
5	$2r + 1$ copies of a generalized quadrangle or its dual; $D = 5$	bijections as in §4; freed sets (partial ovoids and spreads, two-point local dualities, concurrent pairs)	the incidences between copies at the freed points are deleted	hub points and hub lines
6	5α copies of a generalized hexagon, $\alpha = \lfloor (q+3)/2 \rfloor$; $D = 8$ claimed, ≥ 9 as defined	one labelling of the points and lines, the same in every copy (one per copy in the repair); block types; a reading of the figures	none	$10N$ new elements, each incident with the point or the line of one label in each of 2α copies
7	a finite group G (mostly SL$(2, p)$ or PSL$(2, p)$ times a cyclic group); $D = 9, \ldots, 14$	a set S of Δ elements, found by search, mostly closed under an automorphism of order 3	no starting geometry	a point p_g and a line L_g for each $g \in G$; $p_g \mathrel{\mathrm{I}} L_h$ iff $hg^{-1} \in S$

TABLE 1. The constructions at a glance.

So ℓ^* contains X_a and the $q + 1 - |S|$ unmarked points of ℓ, and Λ^* contains λ_a and the $q + 1 - |S|$ lines joining A to them. We call A, ℓ, the X_s and the λ_s the *frame*.

- The construction of Araujo-Pardo and López [1] (*three hats*) uses three marked points: $S = \{0, 1, a\}$, $T = \{0, 1\}$.
- The *four hats* use four: $S = \{0, 1, a, c\}$, $T = \{0, 1, c\}$; this needs $q \geq 3$.

The labels $0, 1, a, c$ are only names. The new elements are, for each $s \in S$, a *hat point* H_s and a *hat line* K_s, and one more point Q and one more line M.

1.2. **The two constructions.** Both constructions follow the same six rules. They differ in two permutations of T: the *links* π, a permutation of T without fixed points, and the *rotation* ρ.

Definition 1.1. The incidence structure $\mathcal{S}$ has as points the points of Π, the point Q and the hat points H_s; as lines the lines of Π, the line M and the hat lines K_s; and exactly the following incidences.

(R1) The incidences of Π, except the flags (X_s, λ_s), $s \in S$; and the flags $(X_t, \lambda_{\rho(t)})$, $t \in T$.

(R2) H_s lies on every line of Π through X_s other than ℓ (q lines, λ_s among them).
(R3) K_s contains every point of λ_s other than A (q points, X_s among them).
(R4) M contains the points of ℓ^* and every hat point H_s.
(R5) Q lies on the lines of Λ^* and on every hat line K_s.
(R6) The *links*: $A \mathrm{I}\, K_a$, $H_a \mathrm{I}\, \ell$, and $H_t \mathrm{I}\, K_{\pi(t)}$ for $t \in T$.

	three hats [1]	four hats
marked points	X_0, X_1, X_a	X_0, X_1, X_a, X_c
flags deleted	(X_a, λ_a)	(X_0, λ_0), (X_c, λ_c), (X_1, λ_1), (X_a, λ_a)
flags added	none	(X_0, λ_c), (X_c, λ_1), (X_1, λ_0)
H_s lies on	the q lines through X_s other than ℓ; M; one link line	
K_s contains	the q points of λ_s other than A; Q; one link point	
M contains	the $q-1$ points of ℓ other than X_0, X_1; H_0, H_1, H_a	the $q-2$ points of ℓ other than X_0, X_1, X_c; H_0, H_1, H_a, H_c
Q lies on	the $q-1$ lines through A other than λ_0, λ_1; K_0, K_1, K_a	the $q-2$ lines through A other than $\lambda_0, \lambda_1, \lambda_c$; K_0, K_1, K_a, K_c
links	$A \mathrm{I}\, K_a$, $H_a \mathrm{I}\, \ell$, $H_0 \mathrm{I}\, K_1$, $H_1 \mathrm{I}\, K_0$	$A \mathrm{I}\, K_a$, $H_a \mathrm{I}\, \ell$, $H_0 \mathrm{I}\, K_c$, $H_c \mathrm{I}\, K_1$, $H_1 \mathrm{I}\, K_0$
π, ρ	$\pi = (0\ 1)$, $\rho = \mathrm{id}$	$\pi = \rho = \sigma = (0\ c\ 1)$: $0 \mapsto c \mapsto 1 \mapsto 0$
new elements	Q, H_0, H_1, H_a; M, K_0, K_1, K_a	Q, H_0, H_1, H_a, H_c; M, K_0, K_1, K_a, K_c

TABLE 2. The two constructions, rule by rule. With $\rho = \mathrm{id}$, rule (R1) deletes and restores the flags (X_t, λ_t), $t \in T$, so the three hats delete only (X_a, λ_a).

So the three hats of [1] are the structure $\mathcal{S}$ built with the labels $S = \{0, 1, a\}$, the links $\pi = (0\ 1)$ and no rotation ($\rho = \mathrm{id}$), and the four hats are $\mathcal{S}$ built with $S = \{0, 1, a, c\}$ and $\pi = \rho = \sigma$. Table 2 puts the two side by side, and Figure 1 shows what rule (R1) and the links do to the flags between ℓ and the lines through A.

Dictionary. In [1] the plane is described from a flag (P, L) (the point and line at infinity); P_0 is a point of L other than P, and L_0 a line through P other than L. In our names $P_0 = A$, $L_0 = \ell$, $(0, s) = X_s$, $[0, s] = \lambda_s$, $Q_0 = Q$, $M_0 = M$, $\widehat{(0, s)} = H_s$, $\widehat{[0, s]} = K_s$. By their Observation 1, the lines through $(0, s)$ other than L_0 are the $[t, s]$ and the points of $[0, s]$ other than P_0 are the (t, s), so their edges $(\widehat{(0, j)}, [t, j])$ and $(\widehat{[0, j]}, (t, j))$ are rules (R2) and (R3). Their edges (P, M_0) and $(M_0, (0, i))$, $i \neq 0, 1$, together say that M contains the points of L_0 other than $(0, 0), (0, 1)$, and with $(M_0, \widehat{(0, j)})$ this is (R4); dually (L, Q_0), $(Q_0, [0, i])$ and $(Q_0, \widehat{[0, j]})$ give (R5). The edges $(P_0, \widehat{[0, a]})$, $(L_0, \widehat{(0, a)})$, $(\widehat{(0, 0)}, \widehat{[0, 1]})$, $(\widehat{(0, 1)}, \widehat{[0, 0]})$ are the links, and the deleted edge $((0, a), [0, a])$ is (R1). The flag (P, L) plays no role in the construction: P is one of the points of ℓ^* and L one of the lines of Λ^*.

1.3. **What the new elements are: a crossed doubling of the frame.** The hat point H_s is a copy of X_s that does not see ℓ; the hat line K_s is a copy of λ_s that does not see A; and (Q, M) is a copy of the anti-flag (A, ℓ) that does not see the elements X_t, λ_t, $t \in T$. This is how [1] prove that the diameter is 3: using their Observation 2 they compare each hat vertex with the vertex it copies. In both constructions one has, more precisely: *H_s is at the same distance from every point and line of Π as X_s is in Π, except from X_s itself and, if $s \neq a$, from ℓ; dually K_s is at the same distance from every element of Π as λ_s, except from λ_s and, if $s \neq a$, from A.* Indeed, once Theorem 1.7 below is proved, any two distinct points are at distance 2 and a point and a line not through it at distance 3, in Π and in $\mathcal{S}$ alike; and by rules (R2) and (R6) the lines of Π through

three hats [1]

	ℓ	λ_0	λ_1	λ_a	M	K_0	K_1	K_a
A		●	●	●				●
X_0	●	●				●		
X_1	●		●				●	
X_a	●			○	●			●
Q				●		●	●	●
H_0		●			●		●	
H_1			●		●	●		
H_a	●			●	●			

four hats

	ℓ	λ_0	λ_c	λ_1	λ_a	M	K_0	K_c	K_1	K_a
A		●	●	●	●					●
X_0	●	○	●				●			
X_c	●		○	●				●		
X_1	●	●		○					●	
X_a	●				○	●				●
Q					●		●	●	●	●
H_0		●				●		●		
H_c			●			●			●	
H_1				●		●	●			
H_a	●				●	●				

TABLE 3. The *gadget*: all incidences among the frame and the new elements (● incident, ○ a flag of Π that was deleted). Every other incidence of a new element is with an element outside the frame, by rules (R2)–(R5). In both tables the block $\{X_t\} \times \{\lambda_t\}$ ($t \in T$) is the permutation matrix of ρ and the block $\{H_t\} \times \{K_t\}$ is that of π: the identity and the transposition (0 1) for the three hats, the 3-cycle σ twice for the four hats.

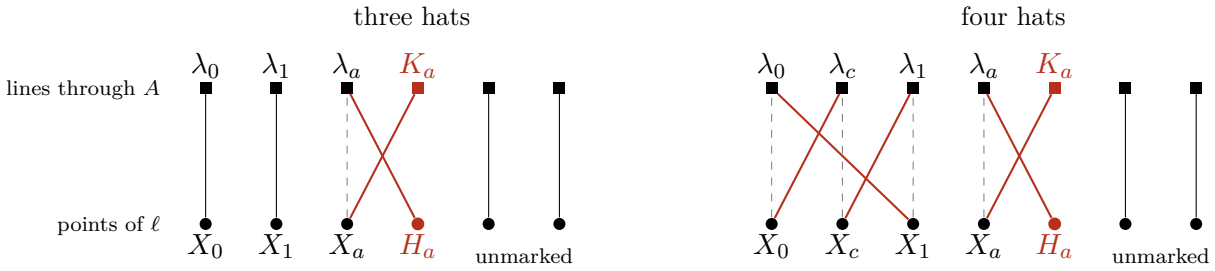


FIGURE 1. The incidences between the points of ℓ and the lines through A (squares are lines, circles points; black: kept flags of Π, dashed: deleted flags, red: new). In Π they form a perfect matching, $X \leftrightarrow AX$. In $\mathcal{S}$ they still form a perfect matching, once the new point H_a (which lies on ℓ) is counted among the points of ℓ and the new line K_a (which passes through A) among the lines through A: X_a is now matched with K_a and λ_a with H_a. The three hats keep the other flags. The four hats replace the three flags (X_t, λ_t), $t \in T$, by the cyclic shift $(X_t, \lambda_{\sigma(t)})$, the *rotation*. There are no other incidences between these points and lines.

H_s are those through X_s other than ℓ, together with ℓ itself if $s = a$. The cleanest way to say all this is as a doubling of the frame, in the sense of Delorme [3], with one twist.

Let $\mathcal{F}$ be the set of frame elements A, ℓ, X_s, λ_s ($s \in S$), and give each $x \in \mathcal{F}$ a copy x' of the same kind:

$$A' = Q, \qquad \ell' = M, \qquad X'_s = H_s, \qquad \lambda'_s = K_s.$$

The *crossed doubling* $\mathcal{D}$ of the frame has the points and lines of Π and these copies, and the following incidences: all incidences of Π; for $x \in \mathcal{F}$ and $y \notin \mathcal{F}$ with $x \mathrel{I} y$, the incidence $x' \mathrel{I} y$ (a copy sees the same elements outside the frame as its original); for $x, y \in \mathcal{F}$ with $x \mathrel{I} y$, the incidence $x' \mathrel{I} y'$ (copies see copies), except that each flag (X_s, λ_s) is *crossed*: instead of $X'_s \mathrel{I} \lambda'_s$ we put $X'_s \mathrel{I} \lambda_s$ and $X_s \mathrel{I} \lambda'_s$.

Reading off the neighbours of the copies: H_s lies on M (as $X_s \mathrel{I} \ell$) and on the lines through X_s other than ℓ (those other than λ_s are outside $\mathcal{F}$; λ_s by the crossing); K_s contains Q and the points

of λ_s other than A; M contains the hat points and the unmarked points of ℓ; Q lies on the hat lines and the lines through A other than the λ_s. Comparing with Definition 1.1:

Proposition 1.2. *$\mathcal{S}$ is obtained from the crossed doubling $\mathcal{D}$ of the frame by*

(i) *adding the* patches $X_a \,\mathrm{I}\, M$, $Q \,\mathrm{I}\, \lambda_a$, $A \,\mathrm{I}\, K_a$, $H_a \,\mathrm{I}\, \ell$ *and* $H_t \,\mathrm{I}\, K_{\pi(t)}$ $(t \in T)$*;*

(ii) *deleting the flags (X_s, λ_s), $s \in S$, and adding the flags $(X_t, \lambda_{\rho(t)})$, $t \in T$ (for the three hats: deleting (X_a, λ_a)).*

So the question "what happens to the combinatorics when these points and lines are added" splits into two: what the doubling does (Lemma 1.3), and what the patches, the deletions and the rotation do (§1.4).

Lemma 1.3. *In the crossed doubling $\mathcal{D}$:*

(a) *X_s and λ_s have degree $q+2$; A, ℓ, Q, M, H_s, K_s have degree $q+1$; an element outside $\mathcal{F}$ has degree $q+2$ if it is on ℓ or on some λ_s (a point), or passes through A or some X_s (a line), and degree $q+1$ otherwise.*

(b) *The pairs of points that are not collinear are exactly*

$$\{H_s, X_t\}\ (s \neq t) \qquad \text{and} \qquad \{H_s, Q\},$$

and also $\{Q, A\}$ if every point of ℓ is marked (the four hats with $q = 3$). Dually, the pairs of lines that are not concurrent are exactly $\{K_s, \lambda_t\}$ $(s \neq t)$ and $\{K_s, M\}$, and also $\{M, \ell\}$ if every point of ℓ is marked.

Proof. (a) X_s gains K_s (crossing) and λ_s gains H_s; A and ℓ gain nothing, since their frame neighbours are copied together with them. Q lies on $|S|$ hat lines and $q+1-|S|$ lines through A; H_s on M and q lines; dually for M, K_s. A point outside $\mathcal{F}$ gains M if it is on ℓ and K_s if it is on λ_s (at most one of these, as it is not A or an X_s).

(b) Two points of Π keep their common line. Q and a point u of Π: if $u = A$, use a line through A outside $\mathcal{F}$ (there is one unless every point of ℓ is marked; then Q and A have no common line, as the hat lines miss A); if Au is not a λ_s, use Au; if $u \neq A$ is on λ_s, use K_s. H_s and a point $u \notin \ell$: use uX_s, which is not ℓ. H_s and X_s: a line through X_s outside $\mathcal{F}$. H_s and an unmarked point of ℓ: M. Two hat points: M. This leaves H_s and X_t $(t \neq s)$, whose only common line in Π is ℓ, which H_s is not on, while $M \not\ni X_t$ and the other lines of H_s meet ℓ only in X_s; and H_s and Q, which have no common line (the lines through H_s are M, which misses Q, and lines of Π through X_s; the only line of Π through X_s and A is λ_s, and $Q \notin \lambda_s$). The statement for lines is the same argument with points and lines exchanged; $\mathcal{D}$ is built from Π and the frame by rules that are symmetric under this exchange. □

So the copies cost nothing among the old points: no two points of Π become non-collinear and no two lines of Π stop meeting. The failures all involve a copy, and they come from paths of length 2 inside the frame: $X_s - \ell - X_t$ (the copy H_s of X_s does not see ℓ, and the original X_t does not see the copy M of ℓ) and $X_s - \lambda_s - A$ (the crossing removes the copy $H_s - K_s - Q$ of this path); the extra pair $\{Q, A\}$ occurs only when every line through A is in the frame, so that A and its copy have no neighbour in common. For Delorme's doubling without crossing, the hat lemma (`POLISHED.md`, Lemma 1.3) says that a set of elements of a projective plane can be doubled without changing the diameter only if each of its connected components is a single element or a flag; the frame contains the paths ℓ, X_s, λ_s, A, so some patching is unavoidable. Lemma 1.3(a) also says that the doubling leaves exactly one free incidence at each of the $2|S|+4$ elements A, ℓ, Q, M, H_s, K_s, and none at X_s, λ_s.

1.4. **The patches, the deletions and the rotation.** The patches of Proposition 1.2(i) spend the free incidences of A, ℓ, Q, M, H_s, K_s, one each. The far pairs of Lemma 1.3(b) are repaired as follows (a pair $\{H_s, X_t\}$ always has $s \neq t$):

far pair in $\mathcal{D}$	common line in $\mathcal{S}$	because of
$\{H_s, X_a\}$	M	patch X_a I M
$\{H_a, X_t\}$	ℓ	patch H_a I ℓ
$\{H_t, X_{\pi(t)}\}$, $t \in T$	$K_{\pi(t)}$	link H_t I $K_{\pi(t)}$ (and $X_{\pi(t)}$ I $K_{\pi(t)}$)
$\{H_t, X_{\sigma^{-1}(t)}\}$, $t \in T$ (four hats)	λ_t	rotation $X_{\sigma^{-1}(t)}$ I λ_t (and H_t I λ_t)
$\{H_a, Q\}$	λ_a	patch Q I λ_a (and H_a I λ_a)
$\{H_t, Q\}$, $t \in T$	$K_{\pi(t)}$	link H_t I $K_{\pi(t)}$
$\{Q, A\}$ (only if every point of ℓ is marked)	λ_a	patch Q I λ_a

The far pairs of lines are repaired dually: $\{K_s, \lambda_a\}$ by Q, $\{K_a, \lambda_t\}$ by A, $\{K_{\pi(t)}, \lambda_t\}$ by H_t, $\{K_t, \lambda_{\sigma(t)}\}$ by X_t (four hats), $\{K_a, M\}$ by X_a, $\{K_{\pi(t)}, M\}$ by H_t, and $\{M, \ell\}$ (only if every point of ℓ is marked) by X_a.

Why the three hats need no rotation and the four hats do. The pairs $\{H_s, X_t\}$ with $s = a$ or $t = a$ are repaired by ℓ and M. What remains is $\{H_t, X_u\}$ for $t \neq u$ in T: each hat point H_t must reach the $|T| - 1$ points X_u, $u \in T \setminus \{t\}$, and it has one free incidence, which goes to a link. For the three hats $|T| - 1 = 1$ and the link H_t I $K_{\pi(t)}$ suffices. For the four hats $|T| - 1 = 2$: the link reaches $X_{\sigma(t)}$, and $X_{\sigma^{-1}(t)}$ is reached through λ_t, which already carries H_t, after the rotation has moved $X_{\sigma^{-1}(t)}$ onto λ_t. Hence $\rho = \pi = \sigma$: $\rho^{-1}(t)$ must be the element of $T \setminus \{t\}$ that π does not reach. (Dually each hat line K_t meets $\lambda_{\sigma(t)}$ in X_t and $\lambda_{\sigma^{-1}(t)}$ in $H_{\sigma^{-1}(t)}$.) With $|T| = 4$ the same count gives a hat point two ways to reach three points, which is one heuristic reason why the pattern stops at four hats; a SAT search confirms that no fifth hat can be added inside this family for $q = 4, 5, 7, 8, 9, 11, 13$ (`research/plus2/REPORT.md`).

Why the flags (X_s, λ_s) are deleted. The patches X_a I M and Q I λ_a raise the degrees of X_a and λ_a, which were already $q + 2$, to $q + 3$; deleting the flag (X_a, λ_a) brings both back to $q + 2$. The rotation is degree-neutral: X_t trades λ_t for $\lambda_{\sigma(t)}$, and λ_t trades X_t for $X_{\sigma^{-1}(t)}$. Deleting a flag (X, λ) from a projective plane makes exactly q pairs of points non-collinear (the pairs $\{X, Y\}$, Y a point of λ other than X) and q pairs of lines non-concurrent (the pairs $\{\lambda, N\}$, N a line through X other than λ). After the deletions and the rotation, the pairs of points of Π that have no common line of Π any more are:

- three hats: $\{X_a, Y\}$ for the q points $Y \neq X_a$ of λ_a, and dually $\{\lambda_a, N\}$ for the q lines $N \neq \lambda_a$ through X_a;
- four hats: $\{X_s, Y\}$ for $s \in S$ and the $q - 1$ points Y of λ_s other than A and X_s, and $\{X_a, A\}$; dually $\{\lambda_s, N\}$ for the $q - 1$ lines N through X_s other than ℓ and λ_s, and $\{\lambda_a, \ell\}$. That is $4(q - 1) + 1$ pairs of each kind. The pairs $\{X_t, A\}$ and $\{\lambda_t, \ell\}$, $t \in T$, are not broken, because X_t and A are both on $\lambda_{\sigma(t)}$, and λ_t and ℓ both contain $X_{\sigma^{-1}(t)}$.

Each broken pair is repaired by the crossing and one patch: X_s and the points of λ_s other than A are all on K_s (rule (R3)), and A I K_a repairs $\{X_a, A\}$; dually λ_s and the lines through X_s other than ℓ all pass through H_s, and H_a I ℓ repairs $\{\lambda_a, \ell\}$. This is why the flags may be deleted at all: the crossing has already given every point of λ_s a second line through X_s (namely K_s), and every line through X_s a second point on λ_s (namely H_s).

Figure 2 shows the local picture at a hat pair; Table 3 shows the whole gadget.

1.5. **Degrees and counts.**

Lemma 1.4. *In $\mathcal{S}$ every element has degree $q+1$ or $q+2$. The points of degree $q+1$ (the* deficient *points) are the points of Π off ℓ, other than A, on the lines joining A to the unmarked points of ℓ; dually, the lines of degree $q+1$ are the lines of Π not through A, other than ℓ, through the unmarked points of ℓ. There are $(q-1)(q+1-|S|)$ of each: $(q-1)(q-2)$ for the three hats and $(q-1)(q-3)$ for the four hats. Each deficient point lies on exactly $q - |S|$ deficient lines, and each deficient line has exactly $q - |S|$ deficient points.*

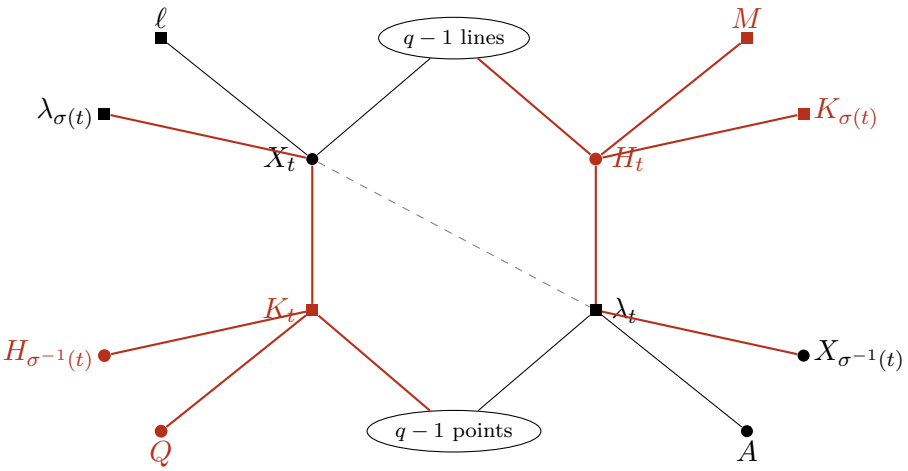


FIGURE 2. The local picture at a hat pair (H_t, K_t), $t \in T$, of the four hats, in the incidence graph. Circles are points, squares are lines, new elements and new incidences are red, the dashed edge is the deleted flag (X_t, λ_t). The upper blob is the set of $q-1$ lines through X_t other than ℓ and λ_t, the lower blob the $q-1$ points of λ_t other than A and X_t. H_t and X_t are twins on the upper blob, K_t and λ_t twins on the lower blob, and the flag (X_t, λ_t) is replaced by the crossed pair X_t I K_t, H_t I λ_t. Every element shown has degree $q+2$. For the three hats ($t \in \{0,1\}$) the dashed flag is kept, the edges to $\lambda_{\sigma(t)}$ and $X_{\sigma^{-1}(t)}$ are absent, and the links go to $K_{\pi(t)}$ and $H_{\pi(t)}$, $\pi = (0\ 1)$.

Proof. Count the lines through each point: those of Π, minus deleted flags, plus added ones. A: $q+1$ lines and K_a. An unmarked point of ℓ: $q+1$ lines and M. X_a: loses λ_a, gains M and K_a. X_t, $t \in T$: four hats, loses λ_t and gains $\lambda_{\sigma(t)}$ and K_t; three hats, gains K_t. A point $p \notin \ell$, $p \neq A$: it is on exactly one line through A, the line Ap; it loses nothing, and gains K_s exactly when $Ap = \lambda_s$. Q: $|\Lambda^*| + |S| = (q+1-|T|) + |S| = q+2$. H_s: q lines by (R2), M, and one link line. The lines are the dual count (Lemma 1.6 below). A deficient point p lies on Ap and on q lines meeting ℓ in the q points other than $Ap \cap \ell$; among these, $q - |S|$ are unmarked. □

So $\mathcal{S}$ has $q^2+q+1+|S|+1$ points and as many lines: $2(q^2+q+5)$ elements for the three hats and $2(q^2+q+6)$ for the four hats. To make it regular, join each deficient point to a deficient line not through it, by a perfect matching. Such a matching exists when there are deficient elements ($|S| \leq q$; for the four hats with $q = 3$ the structure is already regular): the bipartite graph of non-incident pairs between deficient points and deficient lines is regular of degree $(q-1)(q+1-|S|)-(q-|S|) > 0$, and a regular bipartite graph of positive degree has a perfect matching (König). Adding flags cannot increase distances. (In $\mathrm{PG}(2,q)$ explicit matchings are given in coordinates in `coords.tex`, and in [1] up to the index set discussed in Appendix A; any matching of non-incident pairs works.)

1.6. **The diameter.**

Lemma 1.5 (Points)**.** *Any two distinct points of $\mathcal{S}$ are collinear in $\mathcal{S}$; Table 4 gives a common line for every pair.*

Proof. We check that Table 4 covers every pair and that each named line contains both points. The only flags of Π missing from $\mathcal{S}$ are among the (X_s, λ_s); so a flag (u, L) of Π survives whenever $u \notin \ell$ or L does not pass through A.

Two points of Π. If the line uv of Π lost an incidence, then (up to order) $u = X_s$, $uv = \lambda_s$ with the flag deleted, and v is A or another point of λ_s. If $v \neq A$, both lie on K_s by (R3). If $v = A$ and

pair of points	common line	three hats
two points u, v of Π		
the line uv of Π keeps both incidences	uv	
$u = X_s$, v a point of λ_s other than A, X_s, flag (X_s, λ_s) deleted	K_s	only $s = a$
$u = X_a$, $v = A$	K_a	
$u = X_t$, $v = A$, $t \in T$ (four hats)	$\lambda_{\sigma(t)}$	λ_t is kept
Q and a point u of Π		
$u = A$	λ_a	
$u = X_s$	K_s	
u an unmarked point of ℓ	Au	
$u \notin \ell$, $u \neq A$, $Au \in \Lambda^*$	Au	
$u \notin \ell$, $u \neq A$, $Au = \lambda_t$ with $t \in T$	K_t	
H_s and a point u of Π		
$u \notin \ell$ (including $u = A$)	uX_s	
$u = X_s$	a line through X_s other than ℓ, λ_s	
$u \in \ell^*$, $u \neq X_s$	M	
$s = a$, $u = X_t$ with $t \in T$	ℓ	
$s = t \in T$, $u = X_{\pi(t)}$	$K_{\pi(t)}$	$\pi = (0\ 1)$
$s = t \in T$, $u = X_{\sigma^{-1}(t)}$ (four hats)	λ_t	does not occur
two new points		
H_s and $H_{s'}$	M	
Q and H_a	λ_a	
Q and H_t, $t \in T$	$K_{\pi(t)}$	

TABLE 4. A common line for every pair of distinct points of $\mathcal{S}$. Here $s, s' \in S$ and $t \in T$. The third column notes where the three hats differ; elsewhere the entry is the same for both constructions.

$s = a$: $X_a \,\mathrm{I}\, K_a$ by (R3) and $A \,\mathrm{I}\, K_a$ by (R6). If $v = A$ and $s = t \in T$ (four hats): the rotation puts X_t on $\lambda_{\sigma(t)}$, which passes through A.

Q and a point u of Π. $\lambda_a \in \Lambda^*$ passes through A and carries Q. K_s contains X_s and Q. An unmarked point u of ℓ: $Au \in \Lambda^*$, and the flag (u, Au) was not touched. A point $u \notin \ell$, $u \neq A$: the flag (u, Au) survives; if $Au \in \Lambda^*$ it carries Q; otherwise $Au = \lambda_t$, $t \in T$, and u is a point of λ_t other than A, so $u \,\mathrm{I}\, K_t \ni Q$.

H_s and a point u of Π. If $u \notin \ell$, the line uX_s of Π is not ℓ, so it carries H_s, and it contains u in $\mathcal{S}$. If $u = X_s$, any of the $q - 1 \geq 1$ lines through X_s other than ℓ and λ_s carries both. If $u \in \ell^*$ (this includes X_a when $s \neq a$), use M. The remaining case is $u = X_{t'}$ with $t' \in T \setminus \{s\}$. If $s = a$, then $H_a \,\mathrm{I}\, \ell$. If $s = t \in T$: for the three hats $t' = \pi(t)$; for the four hats $t' \in \{\sigma(t), \sigma^{-1}(t)\}$, as σ is a 3-cycle. The line $K_{\pi(t)}$ contains $X_{\pi(t)}$ by (R3) and H_t by the link. The line λ_t contains $X_{\sigma^{-1}(t)}$ by the rotation and H_t by (R2).

Two new points. M carries all hat points. λ_a carries Q (as $\lambda_a \in \Lambda^*$) and H_a (as a line through X_a other than ℓ). $K_{\pi(t)}$ carries Q and, by the link, H_t. □

For lines we use duality. In the dual plane Π^D the line ℓ is a point and A a line not through it, the lines λ_s are points on the line A, and the line joining the points ℓ and λ_s is the point X_s.

Lemma 1.6 (Duality)**.** *The dual of* $\mathcal{S}(\Pi; A, \ell, (X_s), \pi, \rho)$ *is* $\mathcal{S}(\Pi^D; \ell, A, (\lambda_s), \pi^{-1}, \rho^{-1})$*, by the identity on the elements of* Π *and* $H_s \leftrightarrow K_s$*,* $Q \leftrightarrow M$*. Consequently any two distinct lines of* $\mathcal{S}$ *are concurrent.*

Proof. Read the six rules in Π^D with the frame $(\ell, A, (\lambda_s))$. Rule (R1) deletes the flags (λ_s, X_s) and adds $\lambda_t \mathrel{I} X_{\rho^{-1}(t)}$, which is the rotation $X_u \mathrel{I} \lambda_{\rho(u)}$ of $\mathcal{S}$. Rule (R2) gives a new element incident with every element incident with λ_s other than A: this is K_s; rule (R3) gives one incident with every element incident with X_s other than ℓ: this is H_s. Rule (R4) gives a new element incident with the lines through A other than the λ_t, $t \in T$, and with all K_s: this is Q; likewise (R5) gives M. Rule (R6) gives $\ell \mathrel{I} H_a$, $K_a \mathrel{I} A$ and $K_t \mathrel{I} H_{\pi^{-1}(t)}$, which are the links of $\mathcal{S}$. Π^D is a projective plane of order q. For the three hats $\pi^{-1} = \pi$ and $\rho^{-1} = \rho = \mathrm{id}$. For the four hats $\pi^{-1} = \rho^{-1} = \sigma^{-1}$ is again a 3-cycle on T; the proof of Lemma 1.5 uses only that $\pi = \rho$ is a 3-cycle on T (renaming the labels 1 and c turns σ^{-1} into σ). So Lemma 1.5, applied to the dual structure, says that any two lines of $\mathcal{S}$ are concurrent. □

Theorem 1.7. *Let* Π *be a projective plane of order* $q \geq 3$*. The three-hat structure of* [1] *and the four-hat structure are incidence structures with all degrees* $q+1$ *or* $q+2$ *and diameter* 3*, with* $2(q^2+q+5)$ *and* $2(q^2+q+6)$ *elements. Adding a perfect matching of non-incident pairs between the deficient points and lines makes them* $(q+2)$*-regular. Hence* $n_b(q+2, 3) \geq 2(q^2+q+6)$ *for every prime power* $q \geq 3$*.*

Proof. Degrees and counts are Lemma 1.4. Every element is incident with something. By Lemmas 1.5 and 1.6 any two points are collinear and any two lines concurrent, so by Lemma 0.1(a) the diameter is at most 3; it is 3 because $A \not\mathrel{I} \ell$. □

Only the axioms of a projective plane are used, so the theorem holds in non-Desarguesian planes too (we checked the two nearfield planes of order 9). In the range of the tables, $q = 9, 11, 13$ give 192, 276, 376 at $\Delta = 11, 13, 15$, two more than [1].

1.7. **From three hats to four, step by step.** Start from the three hats with labels $0, 1, a$ and choose a fourth point X_c of ℓ, different from X_0, X_1, X_a. The four-hat structure arises by exactly these changes:

(1) add the point H_c, on the q lines through X_c other than ℓ, on M and on K_1; add the line K_c, through the q points of λ_c other than A, through Q and through H_0;
(2) delete the flags (X_0, λ_0), (X_c, λ_c), (X_1, λ_1), and add (X_0, λ_c), (X_c, λ_1), (X_1, λ_0);
(3) delete $X_c \mathrel{I} M$ and $Q \mathrel{I} \lambda_c$ (now X_c is a T-point and λ_c a T-line);
(4) delete the link $H_0 \mathrel{I} K_1$; it is replaced by $H_0 \mathrel{I} K_c$ and $H_c \mathrel{I} K_1$ from (1), while $H_1 \mathrel{I} K_0$ is kept.

In terms of the links, the cycle $0 \to 1 \to 0$ of $\pi = (0\ 1)$ becomes the cycle $0 \to c \to 1 \to 0$ of σ. The four-hat structure has $2q+1$ more flags than the three-hat structure: $2(q+2)$ at H_c and K_c, three from the rotation, minus six. (Checked by computer against the coordinate versions for $q = 4, 5, 7$ and every choice of a, c.)

2. Diameter 4: clones and shadow lines in generalized quadrangles

A *generalized quadrangle* of order (s, t) is an incidence structure in which every line has $s+1$ points, every point is on $t+1$ lines, and

for every anti-flag (p, L) there is *exactly one* point of L collinear with p.

It has $(s+1)(st+1)$ points and $(t+1)(st+1)$ lines, and its incidence graph has diameter 4 and girth 8. By Lemma 0.1(b), an incidence structure has diameter at most 4 as soon as for every anti-flag (p, L) there is *at least one* point of L collinear with p. All constructions in this section only add points, lines and incidences to a quadrangle; nothing is deleted or moved, so no pair of points stops being collinear.

This is the first difference with diameter 3. In a projective plane two cloned points (new points on all lines through an old point) would overload the line joining them, and the hats of Section 1 are clones of the points X_s that avoid the common line ℓ. In a quadrangle, a clone of a point inherits the covering property from its point at once; what has to be arranged is only that every line receives the same number of clones, and this is what an m-ovoid is.

2.1. **Clones and shadow lines.** Let $\mathcal{S}$ be a generalized quadrangle of order (s,t).

- A *clone* of a point p is a new point incident with exactly the lines of $\mathcal{S}$ through p.
- A *shadow line* over a line L is a new line that contains, for each point p of L, exactly one point *over* p: either p itself or one of the clones of p, called the *stand-in* for p.

Definition 2.1 (Clones and shadow lines)**.** Let w assign to every point p a number $w(p) \geq 0$, and let $\mathcal{M}$ be a finite list of lines of $\mathcal{S}$ (repetitions allowed). The structure $\mathcal{S}[w,\mathcal{M}]$ has the points and lines of $\mathcal{S}$, $w(p)$ clones of each point p, and one shadow line over each entry of $\mathcal{M}$, with a chosen stand-in for each of its points. Its incidences are those of $\mathcal{S}$, those of the clones, and those of the shadow lines with their points (Figure 3).

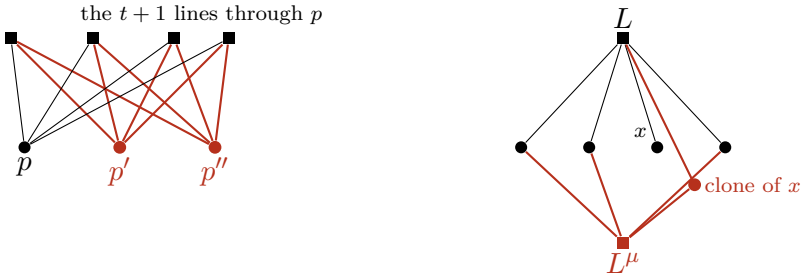


FIGURE 3. Left: a point p with two clones p', p''; all three are on the $t+1$ lines through p and on no other line of $\mathcal{S}$. Right: a shadow line L^μ over L; for the point x of L the stand-in is a clone of x, for the other points of L it is the point itself. The clone of x is on L as well, being on every line through x.

What happens to the combinatorics. No incidence is deleted, so every pair of old points that was collinear stays collinear. A point p and its clones are pairwise collinear, on $t+1$ common lines; two lines through p now meet in $1+w(p)$ points; a shadow line whose stand-ins are all original has all its $s+1$ points on its line (and the same points, if that line carries no clones). So the structure is no longer a partial linear space, but these coincidences are harmless. Two old points that were not collinear stay non-collinear: they have no common old line, a shadow line over L contains only points over points of L, and an old point is over itself only.

Why the diameter is 4. Write $\bar{x}$ for the old point under a point x ($\bar{x} = x$ for an old point, $\bar{x} = p$ for a clone of p) and $\bar{Y}$ for the old line under a line Y. Every point x is on every old line through $\bar{x}$. Let (x, Y) be an anti-flag.

- Y old. Then $\bar{x} \not\mathrm{I} Y$; let r be the point of Y collinear with $\bar{x}$, on the line N. Then $x \mathrm{I} N \mathrm{I} r \mathrm{I} Y$.
- Y a shadow line, $\bar{x} \not\mathrm{I} \bar{Y}$. Let r be the point of $\bar{Y}$ collinear with $\bar{x}$, on the line N, and r^* the stand-in for r on Y. Then $x \mathrm{I} N \mathrm{I} r^* \mathrm{I} Y$, as r^* is on every old line through r.
- Y a shadow line, $\bar{x} \mathrm{I} \bar{Y}$. Take a point $r \neq \bar{x}$ of $\bar{Y}$ and its stand-in r^*: then $x \mathrm{I} \bar{Y} \mathrm{I} r^* \mathrm{I} Y$.

So in every case some point of Y is collinear with x, and the diameter is at most 4 (Lemma 0.1(b)); it is 4 because non-collinear old points remain.

Degrees. An old line L has $s+1+\sum_{p \mathrm{I} L} w(p)$ points. A shadow line has $s+1$ points. An old point or a clone is on $t+1$ old lines plus the shadow lines for which it is the stand-in. This proves:

Theorem 2.2 (Theorem B)**.** *For every generalized quadrangle $\mathcal{S}$ of order (s,t), every w, $\mathcal{M}$ and choice of stand-ins, $\mathcal{S}[w,\mathcal{M}]$ has diameter* 4*, $(s+1)(st+1)+\sum_p w(p)$ points and $(t+1)(st+1)+|\mathcal{M}|$ lines. Its maximum degree is at most Δ if and only if $\Delta \geq \max(s,t)+1$, every line of $\mathcal{S}$ carries clones of total number at most $\Delta-s-1$, and every point (old or clone) is the stand-in for at most $\Delta-t-1$ entries of $\mathcal{M}$.*

2.2. **Exact weights: m-ovoids and covers.** A *weighted m-ovoid* is a weight w on the points such that every line carries total weight exactly m; equivalently, a multiset of points meeting every line in exactly m points, counted with multiplicity. It is an *m-ovoid* if it is a set, and an *ovoid* if moreover $m=1$ (a set of points meeting every line once). Dually, a *weighted m-cover* is a multiset of lines with exactly m members through every point, and a *spread* is a set of lines partitioning the points.

Corollary 2.3. *Let $\Delta \geq \max(s,t)+1$, let w be a weighted $(\Delta-s-1)$-ovoid and $\mathcal{M}$ a weighted $(\Delta-t-1)$-cover, with every stand-in original. Then every old element of $\mathcal{S}[w,\mathcal{M}]$ has degree exactly Δ, every clone $t+1$ and every shadow line $s+1$, and there are $2\Delta(st+1)$ elements in all. Joining the clones to the shadow lines by a suitable set of anti-flags makes the structure Δ-regular, with diameter still* 4*.*

(Counting flags gives $\sum_p w(p)=(\Delta-s-1)(st+1)$ and $|\mathcal{M}|=(\Delta-t-1)(st+1)$. No clone is on a shadow line, so all clone–shadow pairs are anti-flags, and the missing incidences can be added in any balanced way, for instance cyclically; `POLISHED.md`, Corollary 3.3.)

So for diameter 4 the whole question is: *which quadrangles carry weighted m-ovoids and covers for the right m?* Three answers give the lab's values.

2.3. **Theorem A: two elliptic quadrics in the Ahrens–Szekeres quadrangle.** Let q be odd and $\Delta=q+2$. The quadrangle is $\mathrm{AS}(q)$, of order $(q-1,q+1)$; then $\Delta-s-1=2$ and $\Delta-t-1=0$, so we need a *weighted* 2*-ovoid of* $\mathrm{AS}(q)$ *and no shadow lines.* Every point of $\mathrm{AS}(q)$ is already on $t+1=q+2$ lines, so the clones have full degree and the result is $(q+2)$-regular.

The quadrangle. $\mathrm{AS}(q)$ is Payne's derived quadrangle of the symplectic quadrangle $W(q)$ at a point x [7, 8], and its definition is combinatorial. For points u,v write $\{u,v\}^{\perp}$ for the set of points collinear with both, and $\{u,v\}^{\perp\perp}$ for the set of points collinear with every point of $\{u,v\}^{\perp}$; in $W(q)$, for u,v not collinear, $\{u,v\}^{\perp\perp}$ has $q+1$ points (every point of $W(q)$ is regular). Then:

- the points of $\mathrm{AS}(q)$ are the q^3 points of $W(q)$ not collinear with x (and different from x);
- its lines are (i) the lines of $W(q)$ not through x, each without its unique point collinear with x, and (ii) the sets $\{x,y\}^{\perp\perp}\setminus\{x\}$, y not collinear with x (the *hyperbolic lines*).

Every line has q points, every point is on $q+2$ lines, and the q^2 hyperbolic lines form a spread of $\mathrm{AS}(q)$.

The configuration used. Two sets $\mathcal{E}_1,\mathcal{E}_2$ of q^2 points of $\mathrm{AS}(q)$ such that

(E1) each hyperbolic line meets $\mathcal{E}_1$ in exactly one point and $\mathcal{E}_2$ in exactly one point;
(E2) every other line of $\mathrm{AS}(q)$ meets $\mathcal{E}_1$ and $\mathcal{E}_2$ in exactly two points in total (a point of $\mathcal{E}_1\cap\mathcal{E}_2$ counted twice).

Then $w=\mathbf{1}_{\mathcal{E}_1}+\mathbf{1}_{\mathcal{E}_2}$ is a weighted 2-ovoid. Such a pair exists for every odd q: two elliptic quadrics of $\mathrm{PG}(3,q)$ through x, both tangent to the plane of points collinear with x, in a suitable relative position; they meet in x and in one point e of $\mathrm{AS}(q)$ (`POLISHED.md`, Theorem 4.1; the proof is a discriminant computation, and a line of type (i) meets the two quadrics in $1+\chi$ and $1-\chi$ points for some $\chi\in\{-1,0,1\}$). So:

Clone every point of $\mathcal{E}_1$ and every point of $\mathcal{E}_2$ once (so e gets two clones).

Each line of $\mathrm{AS}(q)$ gains exactly two clones, each clone is on the $q+2$ lines of its point, nothing else changes, and Theorem 2.2 gives a $(q+2)$-regular structure of diameter 4 with $q^3+q^2(q+2)+2q^2=2q^2(q+2)$ elements. At $q=9,11,13$: 1782, 3146, 5070. Weights are needed: $\mathrm{AS}(3)$ has no 2-ovoid

and AS(5) no m-ovoid for $m \le 4$ (integer programs, `POLISHED.md` §4). And nothing more can be added at $\Delta = q+2$: points are full, and every line has room for exactly two clones.

2.4. **Proposition C: a 2-ovoid and a spread of $W(q)$.** Let q be odd and $\Delta = q+3$ in $W(q)$, of order (q,q); now $\Delta - s - 1 = \Delta - t - 1 = 2$. The configuration used is a 2-*ovoid* $\mathcal{O}$ of $W(q)$ (a set of $2(q^2+1)$ points meeting every line in exactly two points) and a *spread* Σ (q^2+1 lines partitioning the points). The construction:

Clone every point of $\mathcal{O}$ once, and put two shadow lines over every line of Σ.

Every old line gains two clones, every old point is on two shadow lines (those over its spread line), clones and shadow lines have degree $q+1$, and Corollary 2.3 completes it to a $(q+3)$-regular structure of diameter 4 with $2(q^2+1)(q+3)$ elements; one explicit completion joins the i-th clone to the shadow lines $2i$ and $2i+1$ (indices modulo $2(q^2+1)$, in any numbering). At $q = 13$ this is 5440 at $\Delta = 16$. 2-ovoids of $W(q)$, q odd, exist: the union of two suitable classes of Ebert's partition of $\mathrm{PG}(3,q)$ into elliptic quadrics (orbits of a Singer subgroup) is one (`POLISHED.md`, Proposition 5.1); see also [2]. Spreads of $W(q)$ exist for every q.

2.5. **Even q, and all degrees.** For even q, $W(q)$ has both an ovoid $\mathcal{O}$ and a spread Σ [8]. With j clones of every point of $\mathcal{O}$ and j shadow lines over every line of Σ, Corollary 2.3 gives a $(q+1+j)$-regular structure of diameter 4 with $2(q+1+j)(q^2+1)$ elements. Combining the three cases (ovoid and spread for even q; $j/2$ copies of a 2-ovoid and j copies of a spread for odd q and even j; AS(q) with $(\Delta-q)/2$ copies of the weighted 2-ovoid of Theorem A and $\Delta - q - 2$ copies of the hyperbolic spread for odd q and odd j) gives $n_b(\Delta, 4) \ge 2q^2\Delta$ for every prime power $q \le \Delta - 1$ (`POLISHED.md`, Corollary 5.3).

2.6. **For comparison: Delorme's doubling, and stand-ins.** *Delorme's construction* behind the 2013 values [3] takes a spread Σ of $W(q)$ and a line $v \in \Sigma$. Each line of Σ other than v gets one copy with the same points (a shadow line with all stand-ins original); each point of v gets one copy, which is on the q lines through the point other than v; and v gets three copies, one through the original points of v and two through the copies. All degrees are at most $q+2$, the diameter is 4, and there are $2(q+1)(q^2+1) + q^2 + q + 4$ elements. It is a doubling in Delorme's sense: the copied set is the spread with the points of one of its lines, at two levels. Theorem 2.2 is not a doubling: since a clone is on *every* old line through its point, no condition on the shape of the copied set is needed, and whole m-ovoids can be cloned.

Stand-ins: the $(7,4)$ example. In the quadrangle $Q(4,5)$ (order $(5,5)$) take an ovoid $\mathcal{O}$ and clone each of its 26 points once: every line gains one clone and has degree 7, every old point has degree 6. Shadow lines can now be added as long as every point serves at most one of them: a point off $\mathcal{O}$ has no clone and can be on at most one shadow line, while a point of $\mathcal{O}$ can be on two, standing in for itself on one and letting its clone stand in on the other. An integer program finds 21 such shadow lines (four points of $\mathcal{O}$ lie on the underlying lines of two of them, and stand in for themselves on one and through their clone on the other), giving $2 \cdot 6 \cdot 26 + 26 + 21 = 359$ elements, maximum degree 7 and diameter 4 (`POLISHED.md`, Appendix A). In this family at most $q^2 - q + 2$ shadow lines are possible (`POLISHED.md`, Remark 3.5).

3. Diameter 6: a spread and whiskers in a generalized hexagon

Let Γ be a generalized hexagon of order (q,q): every line has $q+1$ points, every point is on $q+1$ lines, any two elements are at distance at most 6 in the incidence graph, and there are no circuits of length less than 12. It has $N = (q+1)(q^4+q^2+1)$ points and N lines. Elements at distance 6 are *opposite*; two elements at distance less than 6 are joined by a unique shortest path. A point and a line are at distance 1, 3 or 5. In the constructions, Γ is the split Cayley hexagon $H(q)$ [5, 6], but only in the choice of the special configurations (§3.2).

By Lemma 0.1(c) the diameter stays at most 6 if for every anti-flag (p, L) some point of L is within distance 4 of p. Here the new elements come from a doubling, and the argument is by "sheets".

3.1. **Doubling with levels.** This is Delorme's operation [3, §7], in incidence language. Choose sets of elements (points and lines mixed)

$$\mathcal{A}_0 \supseteq \mathcal{A}_1 \supseteq \mathcal{A}_2 \supseteq \cdots,$$

and call $\max\{k : x \in \mathcal{A}_k\}$ the *level* of x (level -1 if $x \notin \mathcal{A}_0$). Fix $m \geq 2$.

Definition 3.1 (Doubling with levels)**.** Replace every element x of level k by m^{k+1} *versions* x^w, one for each word w of length $k+1$ over $\{1, \ldots, m\}$; an element of level -1 keeps one version, itself. Declare $x^w \mathrel{\mathrm{I}} y^u$ if and only if $x \mathrel{\mathrm{I}} y$ in Γ and one of the words w, u is a prefix of the other.

For $m = 2$ and a single level this says: every element x of $\mathcal{A}_0$ gets a twin x'; inside $\mathcal{A}_0$ originals see originals and twins see twins; an element outside $\mathcal{A}_0$ sees both x and x' whenever it saw x. So an outside element gains one incidence for each element of $\mathcal{A}_0$ it is incident with, and an element of $\mathcal{A}_0$ gains nothing. With more levels, a version x^w of level k sees one version of each neighbour of level at most k (the one whose word is the corresponding prefix of w) and all m versions of a neighbour of level $k+1$ extending w.

Sheets. For an infinite word ω, the versions whose words are prefixes of ω form a copy of Γ (every element has exactly one version in it, and any two of these versions have comparable words). Two sheets share exactly the versions indexed by their common prefixes: all elements of level -1, then the versions of level-0 elements if the words begin with the same letter, and so on (Figure 4). So if r is the highest level, the doubled structure is m^{r+1} copies of Γ glued along a tree.

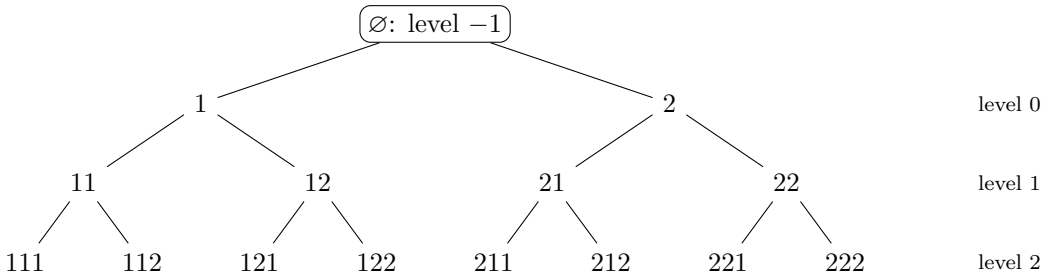


FIGURE 4. Doubling with three levels, $m = 2$. An element of level k has one version for each node at depth $k+1$; versions are incident when their elements are and their nodes lie on a common path from the root. Each leaf (a word of length 3; only the first three letters of an infinite word matter here) is a sheet, a copy of Γ consisting of the versions on the path from the root to that leaf. Elements of level -1 are shared by all eight sheets.

Two conditions control the degrees and one the diameter:

(L0) every element outside $\mathcal{A}_0$ is incident with at most one element of $\mathcal{A}_0$;

(L1) for every $i \geq 0$, every element incident with an element of $\mathcal{A}_{i+1}$ lies in $\mathcal{A}_i$, and every element of $\mathcal{A}_i \setminus \mathcal{A}_{i+1}$ is incident with at most one element of $\mathcal{A}_{i+1}$;

(C) any two elements x, y in one connected component of $\mathcal{A}_0$ (possibly $x = y$) admit an element $z \notin \mathcal{A}_0$ with $d(x, z) + d(z, y) \leq 6$.

Under (L0) and (L1) every version has degree at most $q + 1 + (m - 1)$: it sees one version of each neighbour, except of the one neighbour (if any) one level up, of which it sees m. Under (C) the diameter stays at most 6: two versions in a common sheet are within 6; two versions in no common sheet have incomparable words, so both elements are in $\mathcal{A}_0$; if they lie in different components of

$\mathcal{A}_0$, a shortest path between them in Γ passes through an element of level -1, which belongs to every sheet, and the two halves of the path can be lifted into the two sheets; if they lie in the same component, (C) gives such an element. In a generalized hexagon, (C) holds if and only if every connected component of $\mathcal{A}_0$ is a tree of diameter at most 4 (the hat lemma, `POLISHED.md` Lemma 1.3: a path of length 5 inside $\mathcal{A}_0$ is the unique shortest path between its ends, so no detour through an outside element is short enough). Finally there are $(m-1)\sum_k m^k|\mathcal{A}_k|$ new elements.

Counting flags between $\mathcal{A}_0$ and its complement gives a ceiling: if Γ has order (q,q), (L0) holds, every component of $\mathcal{A}_0$ is a tree and c is the number of components, then $q|\mathcal{A}_0| + 2c \le 2N$, with equality exactly when every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$ (`geometry.tex`, Proposition 3.6).

3.2. **The configuration: a spread, its distinguished line, and whiskers.** *A spread* of Γ is a set Σ of q^3+1 pairwise opposite lines. We use two properties: the lines of Σ have disjoint point sets, so every point of the set $\mathcal{P}_\Sigma$ of points on lines of Σ is on exactly one line of Σ ($|\mathcal{P}_\Sigma| = (q+1)(q^3+1)$); and every line not in Σ meets exactly one line of Σ (a line meeting two lines of Σ would put them at distance 4, and counting gives all N lines). $H(q)$ has spreads for every q: the *Hermitian spreads*, the lines of $H(q)$ in a hyperplane that meets the quadric of $H(q)$ in an elliptic quadric [4].

Fix a line $v \in \Sigma$. The points split by their distance from v: the $q+1$ points of v (distance 1), the $q^2(q+1)$ further points on the $q(q+1)$ lines meeting v (distance 3), and the rest (distance 5). A point of $\mathcal{P}_\Sigma$ not on v is at distance 5 from v, because its line in Σ is opposite v. Call a point *free* if it is not in $\mathcal{P}_\Sigma$ and is at distance 5 from v; there are

$$N - (q+1) - q^2(q+1) - q^3(q+1) = q^3(q^2-1)$$

free points.

Whiskers. Let h be a line opposite v and not in Σ. It meets exactly one line n of Σ, with $n \ne v$, in one point; its other q points are free (they are not in $\mathcal{P}_\Sigma$, since h meets only one line of Σ and only once, and every point of a line opposite v is at distance 5 from v). Such a line is a *whisker*, hanging from the point $h \cap n$.

Definition 3.2. A *whisker set* is a set W of whiskers such that no free point is on two lines of W. It is *exact* if every free point is on exactly one line of W; then $|W| = q^3(q^2-1)/q = q^2(q^2-1)$, the largest possible.

Exact whisker sets exist in $H(q)$ for every q: for odd q an explicit one is cut out by q^2-1 further Hermitian spreads through v (with the spread itself, q^2 spreads pairwise meeting only in v), with one whisker per free point chosen by an affine condition (`POLISHED.md`, Theorem 6.5); for even $q \ge 4$ a characteristic-2 variant (`POLISHED.md`, Theorem 6.14; checked here for $q = 4$); for $q = 2$ one found by computer.

3.3. **The levels, and what each element becomes.** Put

$$\mathcal{A}_2 = \{v\}, \qquad \mathcal{A}_1 = \{v\} \cup \{\text{points of } v\} \cup \Sigma, \qquad \mathcal{A}_0 = \mathcal{A}_1 \cup \mathcal{P}_\Sigma \cup \{\text{lines meeting } v\} \cup W.$$

In words: v is at level 2; the other spread lines and the points of v at level 1; the other points of $\mathcal{P}_\Sigma$, the $q(q+1)$ lines meeting v and the whiskers at level 0; everything else at level -1.

The conditions. (L1): the elements incident with v are its points, which are at level 1, and each sees only v at level 2; the elements incident with a level-1 element are the points of $\mathcal{P}_\Sigma$ and the lines meeting v, all in $\mathcal{A}_0$, and each of them sees exactly one level-1 element (its spread line, or its point on v), while a whisker sees none. (L0) holds, and with W exact it holds with equality: a point at distance 3 from v is on exactly one line meeting v (two would close a circuit of length at most 6) and on no whisker; a free point is on exactly one whisker; a line outside $\mathcal{A}_0$ meets exactly one line of Σ, hence contains exactly one point of $\mathcal{P}_\Sigma$, and no other element of $\mathcal{A}_0$. (C): the connected components of $\mathcal{A}_0$ are

- the *ball* of radius 2 about v: the line v, its $q+1$ points and the $q(q+1)$ lines through them;
- for each of the q^3 lines $n \in \Sigma \setminus \{v\}$, the *broom* of n: the line n, its $q+1$ points, and the whiskers hanging from them.

Both are trees of diameter 4 (Figure 5). Since W is exact, the ceiling $q|\mathcal{A}_0| + 2(q^3+1) = 2N$ is attained: no family of this kind can double more.

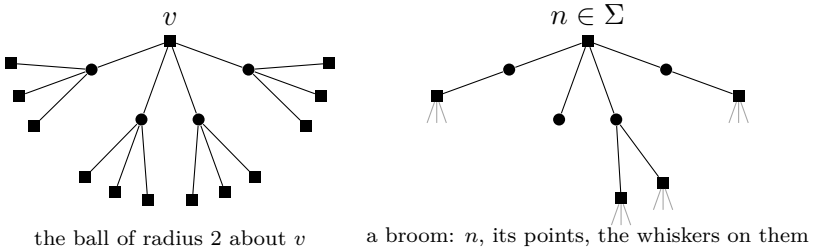


FIGURE 5. The two kinds of connected components of $\mathcal{A}_0$ (squares are lines, circles points; only some of the $q+1$ points and q lines are drawn). Each is a tree of diameter 4. The short grey strokes on a whisker stand for its q free points, which are at level -1. Every element outside $\mathcal{A}_0$ is incident with exactly one element of $\mathcal{A}_0$, so the components together with their neighbours partition the elements of Γ.

Every new element. Write i, j, k for letters in $\{1, \dots, m\}$. The versions and their neighbours are (`POLISHED.md`, §6.2):

element (level)	a version is incident with	degree
v (2): v^{ijk}	p^{ij} for the $q+1$ points p of v	$q+1$
point p of v (1): p^{ij}	$v^{ij1}, \dots, v^{ijm}$; L^i for the q lines $L \neq v$ through p	$q+m$
line $n \in \Sigma \setminus \{v\}$ (1): n^{ij}	p^i for the $q+1$ points p of n	$q+1$
point $p \in \mathcal{P}_\Sigma$ not on v (0): p^i	$n^{i1}, \dots, n^{im}$ for its spread line n; h^i for each whisker h on p; every other line through p	$q+m$
line $L \neq v$ meeting v (0): L^i	$p^{i1}, \dots, p^{im}$ for $p = L \cap v$; the q other points of L	$q+m$
whisker h (0): h^i	p^i for the point p of h in $\mathcal{P}_\Sigma$; the q free points of h	$q+1$
point $r \notin \mathcal{P}_\Sigma$ (-1)	all m versions of the line through r meeting v or of its whisker; every other line through r	$q+m$
line $N \notin \mathcal{A}_0$ (-1)	all m versions of its point in $\mathcal{P}_\Sigma$; its q other points	$q+m$

Nothing is deleted: every flag of Γ survives in every sheet. The elements of level -1 see more (all versions of their one neighbour in $\mathcal{A}_0$), the versions of the spread lines and of the whiskers see less (only versions of their points), and these last are the elements of degree $q+1$.

Theorem 3.3 (Theorem D)**.** *With an exact whisker set, the doubling of $H(q)$ with these levels has maximum degree $q+m$, diameter 6 and*

$$2N + (m-1)\big(2q^4 + 2q^3 + 2q + 2 + m(q^3+q+2) + m^2\big)$$

elements; for $m = 2$ this is $2N + 2q^4 + 4q^3 + 4q + 10$.

At $q = 5, 9, 11, 13$ and $m = 2$ this gives 9592, 148944, 388972, 870440, and at $q = 13$, $m = 3$ it gives 940846. With $W = \emptyset$ the same levels are Delorme's construction [3, §8(b)], which gives the 2013 values; the whiskers add $(m-1)q^2(q^2-1)$.

3.4. **Theorem R: a regular version.** The doubled structure has more lines than points, $(m-1)m(q^3-q+m)$ more, and its deficient elements are the versions of spread lines and whiskers. Two observations make it regular.

- *Repeated lines.* For $n \in \Sigma \setminus \{v\}$ and a letter i, the m versions $n^{i1}, \dots, n^{im}$ have the same points (the p^i, p on n); likewise the versions $v^{ij1}, \dots, v^{ijm}$. Deleting a line that has a twin with the same points changes no distance between the remaining elements (a shortest path through it may use the twin). Delete $(m-1)m(q^3-q+m)$ such lines, at most $m-1$ from each class of m, so that points and lines are equal in number.
- *Completion.* Now every element has degree between $q+1$ and $q+m$, and the deficits on the two sides have the same sum; this sum is large compared with $q+m$, and a swapping argument (`POLISHED.md`, Lemma 7.2) gives a set of anti-flags, one for each missing incidence, whose addition makes the structure $(q+m)$-regular. Adding flags does not increase distances.

The result has $2(q+1)\big(q^4+q^2+1+(m-1)(q^3+m+1)\big)$ elements: 9348, 147500, 386328, 866068 at $q = 5, 9, 11, 13$, $m = 2$, and 927724 at $q = 13$, $m = 3$. The completing anti-flags are not given by a rule; only their existence is proved.

4. Diameter 7: copies of a hexagon joined by matchings, freed sets and hubs

(Theorems A, B, C of this section are those of `D7_HEXAGON.md`, not those of Section 2.)

Let Γ be a generalized hexagon of order (q,q) with N points and N lines, as in Section 3, and Γ^D its dual (points and lines exchanged). Two facts are used repeatedly: a point and a line of Γ are at distance at most 5, two elements of the same kind at most 6; and for any two elements x, y there is an element z at distance 2 from x (a point collinear with x if x is a point, a line concurrent with x if x is a line) with $d(z,y) \le 3$ if x, y are of different kinds and $d(z,y) \le 4$ if they are of the same kind (walk two steps from x along a shortest path to y; if x and y are incident, take for z another element incident with y; if $x = y$, any z).

4.1. **Level structures.** Fix $r \ge 1$ and take $2r+1$ copies $\Gamma_0, \dots, \Gamma_{2r}$ of Γ, indexed modulo $2r+1$. For each copy j and each $k = 1, \dots, r$ (a *transition* from copy j to copy $j+k$) choose a bijection $M_{j,k}$ from the points of Γ_j to the lines of Γ_{j+k}.

Definition 4.1 (Level structure)**.** Its points are the points of all copies, its lines the lines of all copies, and its incidences are those inside each copy together with

$$p \mathrel{\mathrm{I}} M_{j,k}(p) \qquad \text{for every point } p \text{ of } \Gamma_j \text{ and every } k = 1, \dots, r.$$

So every point gains r lines, one in each of the next r copies, and every line gains r points, one from each of the previous r copies: the structure is $(q+1+r)$-regular with $(2r+1)\cdot 2N$ elements. (For each Δ, the largest order $(2r+1)\cdot 2N$ obtainable this way with $\Delta = q+1+r$ is the 2013 table value at diameter 7, which the CombinatoricsWiki table credits to Delorme, Gómez and Quisquater.) A copy may also be a copy of the dual Γ^D; its "points" are then lines of Γ. A transition from a copy of Γ to a copy of Γ^D is called *friendly*: it may join each point p of Γ to p itself, now a line of the dual copy.

Theorem 4.2 (Theorem A of `D7_HEXAGON.md`)**.** *For any choice of the bijections, the level structure has diameter at most* 7*.*

Proof. Two elements of one copy are within 6. Any two copies differ by $\pm k$ with $1 \le k \le r$, since there are $2r+1$ of them; so take u in copy j, v in copy $j+k$, and $M = M_{j,k}$. If u is a point: u, then $M(u)$, then v inside copy $j+k$: at most $1+6$. If u and v are lines: u to the point $M^{-1}(v)$ inside copy j, then v: at most $5+1$. If u is a line and v a point: choose a line b through v; u to the point $M^{-1}(b)$, then b, then v: at most $5+1+1$. □

4.2. **Freeing incidences and adding hubs.** To add elements, incidences must be freed. For each transition $t = (j, k)$ choose a set R_t of points of copy j, the *freed* points, and:

(1) delete the incidences p I $M_t(p)$, $p \in R_t$; each deletion leaves a free incidence (a *slot*) at the point p of copy j and one at the line $M_t(p)$ of copy $j + k$;
(2) add new lines, the *line-hubs*, each incident with one freed point in *every* copy (taking a slot there), and new points, the *point-hubs*, each incident with one freed line in every copy;
(3) add incidences between line-hubs and point-hubs (a *hub graph*), at most $q - r$ at each hub.

All degrees stay at most $q + 1 + r$ (a hub has $2r + 1$ incidences with the copies), and there are $(2r + 1) \cdot 2N$ elements plus the hubs. Figure 6 shows the case $r = 1$.

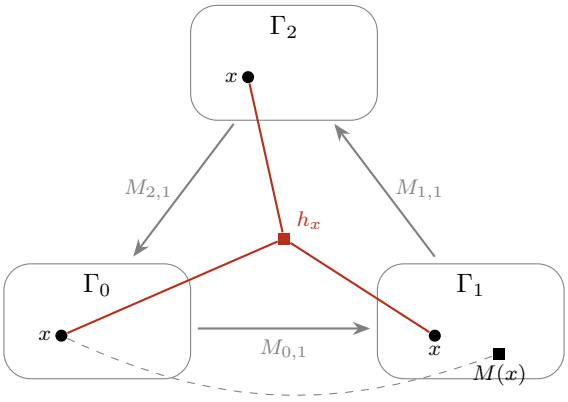


FIGURE 6. Three copies ($r = 1$) joined by matchings M from the points of one copy to the lines of the next. Freeing the point x on the transition $0 \to 1$ deletes the incidence of $x \in \Gamma_0$ with the line $M(x) \in \Gamma_1$ (dashed). In the records with $r = 1$ every transition frees the same set R, and the line-hub h_x takes the slot of the copy of x in each of the three copies; a point-hub g_x (not drawn) takes the slots of the three copies of the line $M(x)$.

What the hubs need. A line-hub is incident with a point in every copy, which is within 6 of everything in its copy: so a hub is within 7 of every element of every copy. A line-hub and a point-hub are within 7 through copy 0 (a point and a line of one copy are within 5). Two hubs of the same kind must be within 6, through the hub graph or through the copies. *What the deletion breaks.* Only pairs in two copies joined by a transition can be affected, and only through that transition. Call a transition *repaired* if after the deletion every element of copy j is still within 7 of every element of copy $j + k$ using only these two copies and the remaining incidences of the transition. Then:

Theorem 4.3 (Assembly, `D7_HEXAGON.md` Theorem 3.4)**.** *If every transition is repaired and any two hubs of the same kind are within distance* 6*, the structure has diameter at most* 7*.*

So everything reduces to a property of one matching and one freed set, and to finding a hub graph. Two answers to the first question:

4.3. **Matchings that extend to an isomorphism: segment sets.** Suppose the transition's bijection is the restriction of an isomorphism ψ from copy j to copy $j+k$ that sends points to lines: the identity of a friendly transition, or a *duality* of Γ (an incidence-preserving bijection exchanging points and lines; $H(q)$ has one exactly when q is a power of 3).

Definition 4.4. A set F of points is a *segment set* if for every $p \in F$ the other points of F collinear with p all lie on one line through p; equivalently, there are no $p, p', p'' \in F$ with p, p' collinear, p', p'' collinear, and p, p'' not collinear (a path of two collinearities that *turns* at p').

Non-collinear sets are segment sets, and so is any subset of the point set of a set of pairwise opposite lines. A segment set has at most $(q+1)(q^3+1)$ points, with equality exactly for the point set of a spread (`D7_HEXAGON.md`, Proposition 1.4: count the lines meeting F in exactly one point).

Theorem 4.5 (Theorem B of `D7_HEXAGON.md`)**.** *Such a transition is repaired if and only if its freed set is a segment set.*

Why. The key fact is that for any two elements u, v of Γ there is a point $p \notin F$ with $d(u,p) + d(p,v) \leq 6$. If some shortest path from u to v has a point outside F, take it. Otherwise all points on all shortest paths lie in F; a shortest path with three consecutive points would turn, so the remaining cases are short (u and v equal, collinear, incident, or at distance 3 or 4), and each is settled by stepping off the line that carries the other points of F (`D7_HEXAGON.md`, Lemma 4.1, seven cases). Now let u be in copy j and v in copy $j+k$, and apply this to u and $\psi^{-1}(v)$ with $F = R_t$: the route u, then p, then the line $\psi(p)$ in copy $j+k$, then v avoids the deleted incidences and has length $d(u,p) + 1 + d(\psi(p), v) = d(u,p) + 1 + d(p, \psi^{-1}(v)) \leq 7$, since ψ preserves distances. Conversely, if $p, p', p'' \in F$ turn at p' and $u \neq m'$ is a line through p'' (m' the line through p' and p''), the only routes of length at most 7 from u in copy j to $\psi(p)$ in copy $j+k$ would cross at p, p' or p'', all freed.

So with a duality ($q = 3^h$) the whole point set of a spread can be freed on every transition, and no larger set.

4.4. **Arbitrary matchings: local dualities.** For a transition whose bijection M (points of copy j to lines of copy $j+k$) is not an isomorphism, it is enough that M looks like a duality near the freed points.

Theorem 4.6 (Theorem C of `D7_HEXAGON.md`)**.** *The transition is repaired if the freed set R satisfies:*

(H1) *no two points of R are collinear;*
(H2) *no two lines of $M(R)$ are concurrent;*
(H3) *for every $x \in R$, M maps the points collinear with x onto the lines concurrent with $M(x)$ (M is a* local duality *at x).*

Proof. Let u be in copy j and v in copy $j+k$; distances are taken inside each copy. (1) u a point not in R: u, $M(u)$, then v: at most $1+6$. (2) u a line, v a point: the $q+1$ lines through v are pairwise concurrent, so by (H2) at most one is in $M(R)$; take another, b, and $a = M^{-1}(b) \notin R$: u to a, then b, then v: at most $5+1+1$. (3) u a line, v a line not in $M(R)$: u to $M^{-1}(v)$, then v: at most $5+1$. (4) u a line, $v = M(x)$ with $x \in R$: there is a point z collinear with x with $d(z,u) \leq 3$; $z \notin R$ by (H1), and $M(z)$ is concurrent with v by (H3): v to $M(z)$, then z, then u: at most $2+1+3$. (5) $u = x \in R$, v a point: there is a line w concurrent with $M(x)$ with $d(w,v) \leq 3$; by (H3) $w = M(z)$ with z collinear with x, and $z \notin R$: x to z, then w, then v: at most $2+1+3$. (6) $u = x \in R$, v a line: the same, with $d(w,v) \leq 4$: at most $2+1+4$. □

Such matchings are easy to build when R is a set of pairwise opposite points (a *partial ovoid*) and $M(R)$ is to be a set of pairwise opposite lines (a *partial spread*) of the same size: then the sets of points collinear with different points of R are disjoint, and so are the sets of lines concurrent with different lines of $M(R)$; send each $x \in R$ to a chosen line, the $q(q+1)$ points collinear with x bijectively onto the $q(q+1)$ lines concurrent with it, and the remaining points arbitrarily onto the remaining lines (`D7_HEXAGON.md`, Proposition 5.2 and Corollary 5.3(a)). More generally R may contain pairs at distance 4 if $M(R)$ has the same pattern of pairs at distance 4 ("mirrored sets"). For example the q^2+1 points of an elliptic quadric section $Q^-(3,q)$ of the quadric of $H(q)$ are pairwise opposite, for every q, and the Hermitian spread supplies pairwise opposite lines.

4.5. **What the records use.** In each case the structure is that of §4.2, every transition is repaired by Theorem 4.5 or 4.6, and the hub graph satisfies the distance condition; the specific sets and hub graphs were found by computer and checked (`D7_HEXAGON.md`, §§8 and 10).

cells	copies	freed sets	elements
$(7,7)$, $(9,7)$, $(10,7)$, $(13,7)$	3 copies of $H(q)$, one matching on all transitions	the same set R on every transition, M a local duality on R (pairwise opposite points plus a few, mirrored); $\lvert R\rvert = 81,295,229,768$	$6N + 2\lvert R\rvert$
$(11,7)$, $(12,7)$	3 or 5 copies of $H(9)$, a duality	the 7300 points of a spread on every transition	$(2r+1)2N + 2\cdot 7300r$
$(14,7)$, $(16,7)$	5 copies, types $HDHDH$	friendly: identity, large non-collinear and non-concurrent sets; others: local dualities	see `D7_HEXAGON.md` §8
$(15,7)$	7 copies, types $HHDHDHD$	friendly: segment sets with some collinear pairs; others: local dualities	

With classical ingredients only (the q^2+1 points of an elliptic quadric section, q^2+1 lines of a Hermitian spread, and an explicit hub graph built from a projective plane with pendant trees) the same theorems give $n_b(q+2,7) \geq 6N + 2(q^2+1)$ for every $q \geq 5$ (`D7_HEXAGON.md`, Corollary 9.1).

5. Diameter 5: level structures of generalized quadrangles, freed sets and hubs

(Theorems A, B, C, E and the other numbered statements cited in this section are those of `D5_QUADRANGLE.md`.)

This section follows the plan of Section 4 one step lower: copies of a generalized quadrangle joined by bijections have diameter 5, and freeing some of the joining incidences makes room for new points and lines. The difference is that at diameter 5 there is an exact criterion for which incidences can be freed, and it is much more restrictive than at diameter 7.

Let $\mathcal{Q}$ be a generalized quadrangle of order (q,q) (Section 2): every line has $q+1$ points, every point is on $q+1$ lines, and for every anti-flag (p,L) exactly one point of L is collinear with p. It has $N = (q+1)(q^2+1)$ points and N lines, and its dual $\mathcal{Q}^D$ (points and lines exchanged) is again a quadrangle of order (q,q). Three facts are used throughout:

- (Q1) two distinct points are collinear or have a common collinear point, so they are at distance 2 or 4; a point and a line are at distance 1 or 3; dually for lines;
- (Q2) a line not through p has exactly one point collinear with p, and a line through p has q of them; dually, a point not on L is on exactly one line concurrent with L;
- (Q3) for every point p and every line L, some point collinear with p lies on L (by (Q2), or another point of L if $p \mathrel{\mathrm{I}} L$); dually.

The quadrangles used are $W(q)$ and $Q(4,q)$ [8], and $Q(4,q)$ is isomorphic to the dual of $W(q)$ [8, 3.2.1]. As in Lemma 0.1, distances in a structure of diameter 5 can be read off from pairs of the same kind:

Lemma 5.1. *An incidence structure in which every element is incident with something has diameter at most* 5 *if and only if any two points, and any two lines, are at distance at most* 4*; that is, any two points are collinear or both collinear with a third point, and dually.*

Proof. Same-kind distances are even and point–line distances odd. If p is a point and L a line, and p' is a point of L, then $d(p,L) \leq d(p,p') + 1 \leq 5$. □

5.1. **Level structures.** Fix $r \geq 1$ and take $2r+1$ copies $\mathcal{Q}_0, \dots, \mathcal{Q}_{2r}$, indexed modulo $2r+1$, each a copy of $\mathcal{Q}$ (*type H*) or of $\mathcal{Q}^D$ (*type D*). For an element x of $\mathcal{Q}$ write x_j for its copy in $\mathcal{Q}_j$; if $\mathcal{Q}_j$ has type D, a line ℓ of $\mathcal{Q}$ gives a point ℓ_j and a point p gives a line p_j. The *transitions* are the pairs $j \to j+k$, $1 \leq k \leq r$; every two copies are joined by exactly one transition. (Any orientation of the pairs in which every copy has r outgoing transitions would do; all layouts below use this one.) For each transition $t = (j \to j')$ choose a bijection M_t from the points of $\mathcal{Q}_j$ to the lines of $\mathcal{Q}_{j'}$.

Definition 5.2 (Level structure)**.** Its points are the points of all copies, its lines the lines of all copies, and its incidences are those inside each copy together with p I $M_t(p)$ for every transition $t = (j \to j')$ and every point p of $\mathcal{Q}_j$.

Every point gains r lines and every line r points, so the structure is $(q+1+r)$-regular with $(2r+1)\cdot 2N$ elements. A transition is *friendly* if the two copies have different types. Then a point p_j of $\mathcal{Q}_j$ and the line $p_{j'}$ of $\mathcal{Q}_{j'}$ come from the same element p of $\mathcal{Q}$, and M_t may be the *identity* $p_j \mapsto p_{j'}$. For $q = 7, 8, 9, 9, 9, 11$ and $2r+1 = 5, 5, 5, 7, 9, 7$ the orders $(2r+1)\cdot 2N$ are the values of the 2013 table in the cells $(10,5)$–$(15,5)$, which the CombinatoricsWiki [11] credits to Delorme, Gómez and Quisquater [9]. We could not find that paper; the cell $(16,5)$, credited to them as well, is not of this form, so we can say only that the orders coincide.

Theorem 5.3 (Theorem A)**.** *For any types and bijections, the level structure has diameter at most* 5*.*

Proof. By Lemma 5.1 it suffices to join two points, and two lines, by a path of length at most 4. Inside one copy this is (Q1). Otherwise take u in $\mathcal{Q}_j$ and v in $\mathcal{Q}_{j'}$ with a transition M from j to j' (one of the two directions is a transition). Two points: u, the line $M(u)$, then v inside $\mathcal{Q}_{j'}$, at most $1+3$. Two lines: u to the point $M^{-1}(v)$ inside $\mathcal{Q}_j$, then v, at most $3+1$. □

5.2. **Freeing incidences and adding hubs.** For each transition $t = (j \to j')$ choose a set R_t of points of $\mathcal{Q}_j$, the *freed* points, and:

(1) delete the incidences p I $M_t(p)$, $p \in R_t$; each deletion leaves a free incidence (a *slot*) at the point p of $\mathcal{Q}_j$ and one at the line $M_t(p)$ of $\mathcal{Q}_{j'}$;
(2) add K new lines, the *line-hubs*, each incident with one freed point in *every* copy (taking a slot there), and K' new points, the *point-hubs*, each incident with one freed line in every copy; every slot is taken at most once;
(3) add incidences between line-hubs and point-hubs (the *hub graph* Y), at most $q-r$ at each hub.

Copy j has $\alpha_j = \sum_{t \text{ out of } j} |R_t|$ free point slots and $\beta_j = \sum_{t \text{ into } j} |R_t|$ free line slots, so $K \le \min_j \alpha_j$ and $K' \le \min_j \beta_j$. All degrees stay at most $q+1+r$ (a hub has $2r+1$ incidences with the copies), and there are $(2r+1)\cdot 2N + K + K'$ elements. This is the operation of §4.2 with quadrangles in place of hexagons.

Call the transition t *repaired* if after the deletion every point of $\mathcal{Q}_j$ is within distance 4 of every point of $\mathcal{Q}_{j'}$, and every line of $\mathcal{Q}_j$ of every line of $\mathcal{Q}_{j'}$, using only these two copies and the remaining incidences of t. A shortest such path uses exactly one incidence of t: it uses an odd number of them, and with three or more it would contain a path between two distinct lines of $\mathcal{Q}_{j'}$ and one between two distinct points of $\mathcal{Q}_j$, each of length at least 2, so its length would be at least $3+2+2=7$. So t is repaired if and only if for all u in $\mathcal{Q}_j$ and v in $\mathcal{Q}_{j'}$ of the same kind there is a point $a \notin R_t$ of $\mathcal{Q}_j$ with

$$(*) \qquad d(u,a) + 1 + d(M_t(a), v) \le 4$$

(distances inside the copies).

Theorem 5.4 (Assembly; Theorems 3.5, 3.6)**.** *If every transition is repaired, any two line-hubs and any two point-hubs are within distance* 4*, and every hub is on at most* $q-r$ *hub incidences, the structure has diameter at most* 5 *and maximum degree at most* $q+1+r$*; so* $n_b(q+1+r, 5) \ge (2r+1)\cdot 2N + K + K'$*.*

More generally (`D5_QUADRANGLE.md`, Theorem 3.6), instead of "repaired" it is enough that for every transition, any two elements of the same kind, one in each of its two copies, are within distance 4 in the whole structure.

Proof. By Lemma 5.1. Two old points or two old lines: inside a copy by (Q1), in different copies through their transition. A line-hub h and an old line v of $\mathcal{Q}_j$: h contains a point of $\mathcal{Q}_j$, which is within 3 of v; dually for point-hubs and old points. Two hubs of the same kind: by hypothesis. □

The hypothesis on the hubs holds, for instance, if Y alone has all same-kind distances at most 4 (§5.8). The hypothesis on the transitions is a property of one bijection and one freed set, and the next theorem decides it exactly.

5.3. **The exact criterion.**

Theorem 5.5 (Theorem E)**.** *Let $t = (j \to j')$ be a transition with bijection M and freed set R. Then t is repaired if and only if for every $x \in R$:*

(E2) *every point of $\mathcal{Q}_{j'}$ lies on $M(a)$ for some point $a \notin R$ collinear with x;*
(E3) *every line of $\mathcal{Q}_j$ contains a point $a \notin R$ such that $M(a)$ and $M(x)$ are concurrent.*

Proof. We check $(*)$. Distances between points of one copy are even, between a point and a line odd. *Two points $u \in \mathcal{Q}_j$, $v \in \mathcal{Q}_{j'}$*: we need $d(u,a) + d(M(a),v) \le 3$, with $d(u,a) \in \{0,2,4\}$ and $d(M(a),v) \in \{1,3\}$. If $u \notin R$, take $a = u$: $d(M(u),v) \le 3$ by (Q1). If $u = x \in R$, then $a \ne x$, so we need $d(x,a) = 2$ and $d(M(a),v) = 1$: a point $a \notin R$ collinear with x with v on $M(a)$. For all v this is (E2). *Two lines $u \in \mathcal{Q}_j$, $v \in \mathcal{Q}_{j'}$*: now $d(u,a) \in \{1,3\}$ and $d(M(a),v) \in \{0,2,4\}$, so the possibilities are $(1,0)$, $(1,2)$, $(3,0)$. If $v = M(b)$ with $b \notin R$, take $a = b$: $d(u,b) \le 3$. If $v = M(x)$ with $x \in R$, then $M(a) = v$ is impossible, so we need a I u, $a \notin R$, and $M(a)$ concurrent with $M(x)$. For all u this is (E3). □

(The note's proof also goes through point–line pairs and a condition (E1), which (E2) implies; with Lemma 5.1 these are not needed. The program `check_d5.py` compares (E2), (E3) with a direct computation of distances on 416 transitions in $W(q)$ and $Q(4,q)$, $q = 2,3$, among them transitions that fail only one of the two.)

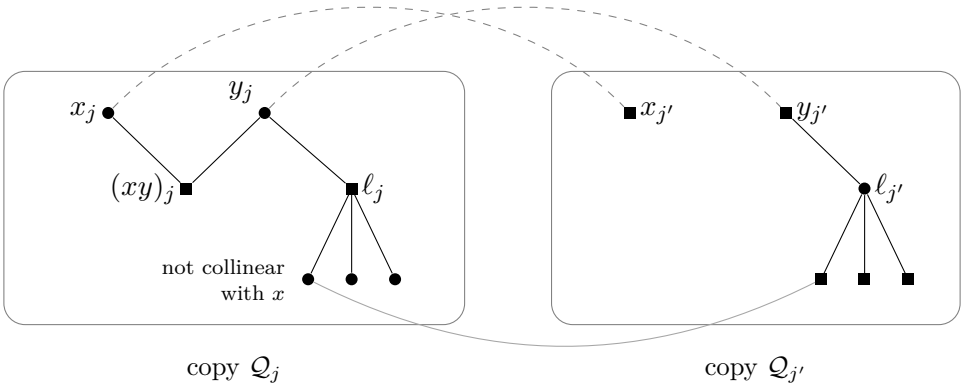


FIGURE 7. Why a friendly transition with M the identity ($\mathcal{Q}_j$ a copy of $\mathcal{Q}$, $\mathcal{Q}_{j'}$ of $\mathcal{Q}^D$) cannot free two collinear points x, y. Dashed: the deleted incidences x_j I $x_{j'}$, y_j I $y_{j'}$. Take a line $\ell \ne xy$ through y; in $\mathcal{Q}_{j'}$ it is a point $\ell_{j'}$. A path of length 4 from x_j to $\ell_{j'}$ must be x_j, a line, a point a_j collinear with x, the line $a_{j'}$, then $\ell_{j'}$; so a must be a point of ℓ collinear with x. The only one is y (Q2), and y is freed. The other points of ℓ (bottom) are not collinear with x, so their incidences with $\mathcal{Q}_{j'}$ (grey) do not help: in these two copies $d(x_j, \ell_{j'}) = 6$.

5.4. **Transitions given by an isomorphism: partial ovoids.** Call a transition *automorphic* if M is the restriction of an isomorphism from $\mathcal{Q}_j$ to the dual of $\mathcal{Q}_{j'}$: the identity of a friendly transition, or a duality of $\mathcal{Q}$ (an incidence-preserving bijection exchanging points and lines; $W(q)$ has one exactly when q is even [8, 3.2.1]).

Theorem 5.6 (Theorem B)**.** *An automorphic transition is repaired if and only if its freed points are pairwise non-collinear (a* partial ovoid *of $\mathcal{Q}_j$).*

Proof. An isomorphism preserves incidence and collinearity, so (E2) and (E3) for x both say: every line of $\mathcal{Q}_j$ contains a point $a \notin R$ collinear with x. A line through x has q points collinear with x, a line ℓ not through x exactly one, p_ℓ (Q2). If no freed point is collinear with x, every line is covered. If a freed point y is collinear with x, every line $\ell \neq xy$ through y has $p_\ell = y$ and is not covered (Figure 7). □

At diameter 7 the corresponding sets were the segment sets (Theorem 4.5), which may contain collinear pairs; at diameter 5 a single collinear pair breaks the transition. In $W(q)$ and $Q(4,q)$ a partial ovoid has at most q^2+1 points, with equality exactly for an *ovoid*, a set of q^2+1 points meeting every line once. $Q(4,q)$ has ovoids for every q, $W(q)$ only for q even [8, 3.4.1, 3.4.2]; for q odd a partial ovoid of $W(q)$ has at most q^2-q+1 points [17]. Dually, freed lines on a friendly transition from a copy of $\mathcal{Q}^D$ must be pairwise non-concurrent lines of $\mathcal{Q}$ (a *partial spread*).

5.5. **Local dualities, and why they stop at two points for odd q.** For a transition between two copies of the same type, M goes from the points to the lines of $\mathcal{Q}$ itself (from the lines to the points, if both copies have type D), and for $Q(4,q)$ with q odd there is no duality. As at diameter 7 (§4.4), it is enough that M looks like a duality near the freed points.

Theorem 5.7 (Theorem C)**.** *The transition is repaired if*

(H1) *no two freed points are collinear, and*

(H3) *for every $x \in R$, M maps the points collinear with x onto the lines concurrent with $M(x)$.*

Proof. (E2): by (H1) no point collinear with x is freed, so by (H3) the lines $M(a)$, $a \notin R$ collinear with x, are all the lines concurrent with $M(x)$, and every point lies on one of them (Q3). (E3): the points a with $M(a)$ concurrent with $M(x)$ are exactly the points collinear with x, none of them freed, and every line contains one (Q3). □

The diameter-7 condition (H2), that the lines $M(R)$ be pairwise non-concurrent, follows here: for $x \neq y$ in R, M maps the $q+1$ points collinear with both onto the lines concurrent with both $M(x)$ and $M(y)$, and two concurrent lines have only $q-1$ such lines (those through their common point).

Two points always work (`D5_QUADRANGLE.md`, Proposition 6.4). Let x, y be non-collinear points and ℓ, m non-concurrent lines. The sets $\{x,y\}$, $\{x,y\}^\perp$ (the $q+1$ points collinear with both), the q^2-1 other points collinear with x, and the q^2-1 other points collinear with y are disjoint, and the same holds for $\{\ell, m\}$ and the lines concurrent with them, with the same sizes. A bijection sending $x \mapsto \ell$, $y \mapsto m$ and each piece onto the corresponding piece, extended arbitrarily, satisfies (H1) and (H3) for $R = \{x,y\}$.

Not three (`D5_QUADRANGLE.md`, Corollary 6.2). Three pairwise non-collinear points form a *triad*, and its *centres* are the points collinear with all three; dually for lines. Let $\{x,y,z\} \subseteq R$ be a triad. By the remark above $M(x), M(y), M(z)$ are pairwise non-concurrent, so they form a triad of lines, and by (H3) M maps the centres of $\{x,y,z\}$ onto the centres of this triad (the lines concurrent with all three). So a triad and its image have the same number of centres. For q odd the points of $Q(4,q)$ are *antiregular* (every triad has 0 or 2 centres) and the points of $W(q)$ *regular* (every triad has 1 or $q+1$ centres) [8, 1.3.6, 3.3.1]. A triad of lines of $Q(4,q)$ is a triad of points of $W(q)$, so the two counts never agree, and a local duality of $Q(4,q)$ (or of $W(q)$) has at most two freed points. At diameter 7 local dualities carried 81 to 768 freed points per transition; here they carry two.

5.6. **Freeing concurrent lines in pairs, and the hub rule.** For a friendly transition to be repaired, its freed set must be a partial ovoid or a partial spread (Theorem 5.6), and in the layouts of §5.7 the number of freed lines on the transitions from type D to type H limits everything. Some concurrent pairs can be freed if the hubs are placed with care.

Let $\mathcal{Q} = Q(4, q)$, let $\mathcal{Q}_j$ have type D and $\mathcal{Q}_{j'}$ type H, and let M be the identity: the point ℓ_j of $\mathcal{Q}_j$ (ℓ a line of $\mathcal{Q}$) is matched with the line $\ell_{j'}$. Free a set T of lines of $\mathcal{Q}$ in which every line meets at most one other line of T, its *partner*.

Proposition 5.8 (Proposition 7.1)**.** *After the deletion, the pairs of the same kind, one element in $\mathcal{Q}_j$ and one in $\mathcal{Q}_{j'}$, that are at distance more than 4 using only these two copies and the remaining incidences of t are exactly the following, for every partner pair ℓ, m with common point z and every point $p \neq z$ of m: the points ℓ_j and $p_{j'}$; the lines p_j and $\ell_{j'}$; and the same with ℓ and m exchanged. That is $4q$ pairs for each partner pair.*

Proof. For a line $\ell \in T$ without partner, (E2) and (E3) hold as in Theorem 5.6. Let ℓ have partner m. (E2) for ℓ asks that every point v of $\mathcal{Q}$ lie on a line $a \notin T$ meeting ℓ, $a \neq \ell$. A point v on ℓ is on q such lines, at most one of them in T. A point v off ℓ is on exactly one line meeting ℓ (Q2), and this line is in T only if it is m, that is, $v \in m \setminus \{z\}$. (E3) for ℓ asks the same of every point u of $\mathcal{Q}$, read as the line u_j of $\mathcal{Q}_j$. Following the proof of Theorem 5.5, the failures are exactly the stated pairs. □

Proposition 5.9 (Proposition 7.2: the hub rule)**.** *Suppose that, for every line $\ell \in T$ with partner m, the slot at the line $\ell_{j'}$ is taken by a point-hub g_ℓ whose line in $\mathcal{Q}_j$ is o_j for some point o of m. Then all the pairs of Proposition 5.8 are within distance* 4.

Proof. For $p \in m \setminus \{z\}$: $\ell_{j'}$, g_ℓ, o_j, m_j, p_j is a walk, since o and p are points of m, so that the lines o_j and p_j of $\mathcal{Q}_j$ pass through its point m_j. And ℓ_j, o'_j, g_m, $m_{j'}$, $p_{j'}$ is a walk, where o' is the point of ℓ at which the hub g_m of the partner sits in $\mathcal{Q}_j$. Both have length 4 (the first is shorter if $o = p$). □

So a transition with partner pairs is not repaired, but with the hub rule it satisfies the weaker hypothesis stated after Theorem 5.4.

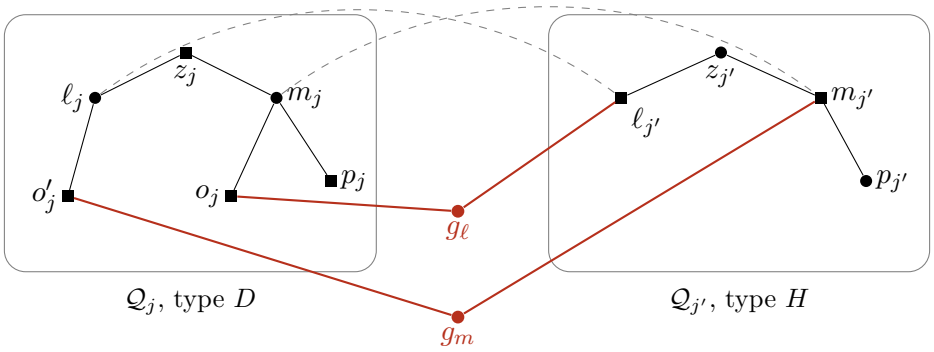


FIGURE 8. The hub rule at a partner pair ℓ, m of freed lines of $\mathcal{Q} = Q(4, q)$ meeting in z. In the copy $\mathcal{Q}_j$ of the dual the lines ℓ, m are collinear points ℓ_j, m_j, and their matched lines $\ell_{j'}, m_{j'}$ (dashed incidences deleted) are concurrent in $\mathcal{Q}_{j'}$. The point-hub g_ℓ that takes the slot at $\ell_{j'}$ is put, in $\mathcal{Q}_j$, on a line o_j through m_j (o a point of m), and likewise g_m on a line o'_j through ℓ_j. Then p_j and $\ell_{j'}$ are joined through g_ℓ, and ℓ_j and $p_{j'}$ through g_m, for every point $p \neq z$ of m; these are the pairs that the deletion broke.

The hubs exist (`D5_QUADRANGLE.md`, Lemma 7.3). In the seven-copy layout of §5.7 the transitions from type H to type D free an ovoid O of $Q(4,q)$, which meets every line exactly once; put g_ℓ, in the two copies of type D, at the point $o = O \cap m$ of the partner. A point of O has three free slots in one of these copies and two in the other, and at most two lines of T pass through a point, so at most two hubs ask for it. The requests can be met whatever the common points of the partners are, even when a partner pair meets at a point of O (then both hubs sit there). One point-hub per partnered line of T suffices (so K' must be at least the number of partnered lines), because in this layout the transitions out of the two copies of type D end in different copies.

How large can T be? Read in $W(q)$, T is a set of points each collinear with at most one other. A single point uses its $q+1$ lines, a pair $2q+1$ lines, and no line is shared, so $|T| \le 2(q+1)(q^2+1)/(2q+1)$, which is 86 at $q=9$; for partial ovoids of $W(q)$, q odd, Tallini's bound is q^2-q+1 [17]. Computer searches found 66 at $q=9$ and 89 at $q=11$ (`D5_QUADRANGLE.md`, §7).

5.7. **Layouts and slot counts.** Over $\mathcal{Q} = Q(4,q)$, q odd, let every transition from type H to type D be the identity and free a points, every transition from D to H the identity and free b lines, and every transition between copies of the same type free c points with a local duality ($c=2$); we assume $a \ge b \ge c$. Counting the transitions out of and into each copy (`D5_QUADRANGLE.md`, Propositions 8.1–8.3):

- five copies $HHDHD$ ($r=2$): $\min_j \alpha_j = \min_j \beta_j = b+c$;
- seven copies $HHHDHHD$ ($r=3$): $\alpha = (a+2c, a+2c, a+2c, 2b+c, a+2c, a+2c, 3b)$, $\beta = (b+2c, b+2c, b+2c, 3a, b+2c, b+2c, 2a+c)$;
- nine copies $HHHHDHHHD$ ($r=4$): $\min_j \alpha_j = \min(a+3c, 3b+c)$, $\min_j \beta_j = b+3c$.

(Over all regular tournaments on seven vertices and all typings, the best value of $\min\alpha + \min\beta$ is $\min(a+b+4c, 3b+3c)$ for $a \ge b \ge c \ge 0$, `D5_QUADRANGLE.md`, §8.) With three copies some copy has no friendly outgoing transition, and hubs are limited by c. For q even all copies have type H over $W(q)$, every transition is a duality restricted to the points, and every transition frees an ovoid of $W(q)$, so $\alpha_j = \beta_j = r(q^2+1)$.

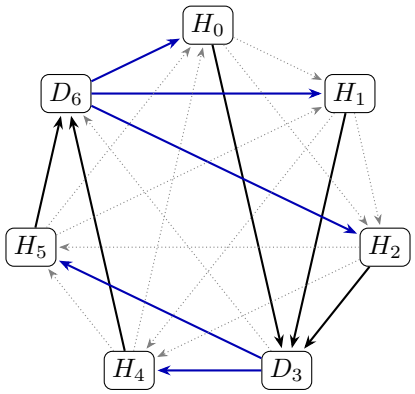


FIGURE 9. The seven-copy layout $HHHDHHD$ (transitions $j \to j+1, j+2, j+3$). Black: the five transitions from type H to type D, identity, freeing an ovoid of $Q(4,q)$ ($a = q^2+1$ points). Blue: the five from D to H, identity, freeing b lines (a partial spread, or a set with partner pairs and the hub rule). Dotted: the eleven between copies of the same type (ten $H \to H$ and $3 \to 6$), each a local duality on two points.

5.8. **Hub graphs.** The hub incidences form a bipartite graph Y whose two sides are the line-hubs and the point-hubs. It needs all degrees at most $\delta = q - r$ and any two vertices of the same side within distance 4. Explicit ones (`D5_QUADRANGLE.md`, Lemma 9.1), either side playing either role, with $N(s) = (s+1)(s^2+1)$:

(Y0) $K, K' \le \delta$: the complete bipartite graph;
(Y1) the incidence graph of a projective plane of order s, $s+1 \le \delta$, with up to $\delta - s - 1$ new vertices of degree one attached to every vertex: $s^2+s+1 \le K, K' \le (s^2+s+1)(\delta - s)$. Two old vertices of one side are at distance 2. A new vertex on c and an old vertex x of its side are within $1+3$, since c and x are on different sides; two new vertices of one side hang on vertices at distance at most 2;
(Y2) the incidence graph of a quadrangle of order (s,s), $s+1 \le \delta$, with up to $\delta - s - 1$ new vertices of degree one attached to each of the $s+1$ lines through a fixed point and to each of the $s+1$ points of a fixed line: $N(s) \le K, K' \le N(s) + (s+1)(\delta - s - 1)$;
(Y3) the incidence graph of the level structure of Theorem 5.3 built on a quadrangle of order (s,s) with $2r'+1$ copies, $s+1+r' \le \delta$: $K = K' = (2r'+1)N(s)$.

5.9. **What the constructions give.** *The ingredients, and why they exist.* An ovoid of $Q(4,q)$: q^2+1 points meeting every line once; it exists for every q [8, 3.4.1], and `D5_QUADRANGLE.md`, Lemma 1.1, gives one explicitly. For q odd, $q \ge 5$, a partial spread of $Q(4,q)$ with $3q-1$ lines, that is, $3q-1$ pairwise non-collinear points of $W(q)$: explicit in `D5_QUADRANGLE.md`, Lemma 1.3 (found by the lab; we did not find it in the sources we read). Two-point local dualities (§5.5). For q even, a duality and an ovoid of $W(q)$ [8, 3.2.1, 3.4.1]. The hub graphs of §5.8.

The families (`D5_QUADRANGLE.md`, Corollaries 10.1–10.4). For q even and $1 \le r \le q-1$: $n_b(q+1+r, 5) \ge (2r+1)\cdot 2N + K + K'$ with $K, K' \le r(q^2+1)$ and a hub graph with $\delta = q - r$. For q odd, $q \ge 5$, with $a = q^2+1$, $b = 3q-1$, $c = 2$ (on the transition between the two copies of type D, $2 \to 4$, $3 \to 6$ and $4 \to 8$ in the three layouts, M goes from the lines to the points of $Q(4,q)$, and the dual of the two-point recipe is used):

$$\begin{array}{lll} n_b(q+3,5) \ge 10N + K + K', & K, K' \le 3q+1, & \delta = q-2; \\ n_b(q+4,5) \ge 14N + K + K', & K \le 6q,\ K' \le 3q+3, & \delta = q-3; \\ n_b(q+5,5) \ge 18N + K + K', & K \le \min(q^2+7, 9q-1),\ K' \le 3q+5, & \delta = q-4. \end{array}$$

Every transition is repaired by Theorems 5.6 and 5.7, and Theorem 5.4 gives the bound. With the hub graphs of §5.8:

cell	2013 table	these families	from	lab value (construction)
$(10,5)$	4,000	4,044	$q=7$, 5 copies, $22+22$ hubs	4,800 (lift of $W(7)$)
$(11,5)$	5,850	6,090	$q=8$, 5 copies, $120+120$	7,020 (lift of $W(8)$)
$(12,5)$	8,200	8,430	$q=8$, 7 copies, $120+120$	9,840 (lift of $W(9)$)
$(13,5)$	11,480	11,552	$q=9$, 7 copies, $42+30$	11,700 (lift of $W(8)$)
$(14,5)$	14,760	14,812	$q=9$, 9 copies, $26+26$	17,568 (lift of $W(11)$)
$(15,5)$	20,496	20,598	$q=11$, 7 copies, $66+36$	20,736 (route-chart lift)

These are above the 2013 table in all six cells and below the lab's values in all six (`D5_QUADRANGLE.md`, §14). With computer-found freed sets the same construction gives 11,636 at $(13,5)$ and 20,715 at $(15,5)$ (seven copies over $Q(4,9)$ and $Q(4,11)$, freed lines with 30 and 36 partner pairs and the hub rule), both checked by breadth-first search from every vertex, and both below the lab's values. A count of slots (`D5_QUADRANGLE.md`, Proposition 14.1) shows that, with at most two freed points on each transition between copies of the same type, no structure of this kind reaches the lab's values in the cells $(10,5)$–$(14,5)$ and $(16,5)$; at $(15,5)$, with an ovoid on the transitions from type H to type D, it would need a set T of at least 111 points of $W(11)$, against the 89 found.

Credits. The level structures themselves are not ours (their orders are the table values credited to [9]). Deleting the edges of a type-preserving matching between copies at a set of pairwise non-collinear points and attaching new vertices is in Rajiv's preprint [13] for general (non-bipartite) graphs; the 'if' half of Theorem 5.6 for friendly transitions is the bipartite version of that step at

diameter 5, with one-sided hubs. Delorme [3, §3] builds cyclic multi-level graphs from a duality of a quadrangle (general graphs); we claim none of this. We did not search the literature for a criterion like Theorem 5.5, for the converse in Theorem 5.6, or for the bound of two points for local dualities.

6. Diameter 8: Gómez's construction on generalized hexagons, and a labelling for each copy

(Theorems 1–4 and the other numbered statements cited in this section are those of `OMEGA_NOTE.md`.)

Six entries of the 2013 table at diameter 8, the cells $(10, 8)$, $(11, 8)$, $(12, 8)$, $(14, 8)$, $(15, 8)$ and $(16, 8)$, are the orders of the compound graphs ΩH_q of J. Gómez [10, §2.9.3] for $q = 7, 8, 9, 11, 11, 13$; the table credits them to Delorme, Gómez and Quisquater [9]. *The construction is Gómez's*: the copies of the hexagon, the wiring of the new elements (his Figures 2.21a,b), the order, the degree, and the claim that the diameter is at most $D_1 + 2 = 8$, where $D_1 = 6$ is the diameter of the hexagon, which the thesis states without proof. *What follows is ours*: the construction in incidence terms; a proof that, as defined, it has diameter at least 9 (§6.2); and the attempt to *repair* it by giving every copy its own labelling (§6.3). The repair works at $q = 2$ and $q = 3$, with labellings found by computer. For every $q \geq 7$ it provably fails for all labellings that respect the geometry (by hand for $q \geq 10$, with computer counts at $q = 7, 8, 9$). For $q = 4, 5$, and for labellings that do not respect the geometry, it is open. So, unlike Sections 1–5, this section describes a construction that does not reach the diameter claimed for it, and why.

Let Γ be a generalized hexagon of order (q, q), with $N = (q + 1)(q^4 + q^2 + 1)$ points and N lines (Section 3); for the table, Γ is the split Cayley hexagon $H(q)$ [5, 6]. Recall: any two elements are at distance at most 6, there are no circuits of length less than 12, and elements at distance 6 are *opposite*. By the argument of Lemma 0.1, an incidence structure has diameter at most 8 if and only if every point is within distance 7 of every line. Two counts in Γ are used.

Lemma 6.1 (Lemmas 1.4 and 1.6)**.** *Let t be an element of Γ.*

(a) *For $1 \leq i \leq 5$ there are $(q + 1)q^{i-1}$ elements at distance i from t, and q^5 elements opposite t.*

(b) *Let z be an element of the same kind as t. Among the elements opposite t, exactly q^4 are within distance 4 of z if $z \neq t$, and none if $z = t$; exactly q^2 are within distance 2 of z if $d(t, z) \in \{4, 6\}$, and none if $d(t, z) \leq 2$.*

Proof. (a) A walk that never steps straight back is a shortest path as long as its length is at most 6, and it is the only shortest path between its ends when its length is at most 5; otherwise two of them would close a circuit shorter than 12. Such walks have $q + 1$ choices for the first step and q for each later one; the remaining $2N - 1 - \sum_{i \leq 5}(q + 1)q^{i-1} = q^5$ elements are opposite t. (b) If $d(t, u) = k$ with $1 \leq k \leq 5$, exactly one element incident with u is at distance $k - 1$ from t and the other q at distance $k + 1$; if $k = 6$, all $q + 1$ are at distance 5. Follow the distance to t along the walks of length at most 4 from z that never step straight back. For example, if $d(t, z) = 2$, a walk of length 4 ends opposite t exactly when every step goes away from t, which happens for $q \cdot q^3$ walks, and the walks of length 2 or 0 end at distance at most 4 from t. The cases $d(t, z) = 4$ and 6 are the same bookkeeping (`OMEGA_NOTE.md`, Lemma 1.6). □

6.1. **Gómez's construction.** Put $\alpha = \lfloor (q+3)/2 \rfloor$, so that 2α is $q+3$ or $q+2$. Take 5α copies Γ_c of Γ, numbered $c = 5B + i$ with a *block* $B \in \{0, \ldots, \alpha - 1\}$ and a *position* $i \in \mathbb{Z}_5$; each block has a *type*, a or b. The thesis gives the points of Γ a set of N labels, and the lines the same set of labels, and uses *the same labels in every copy*; write $P_0(t)$ and $L_0(t)$ for the point and the line with label t. We write y_c for the copy in Γ_c of an element y of Γ.

The new elements come in ten *classes*: an *unbarred* class k and a *barred* class $\bar{k}$ for each $k \in \mathbb{Z}_5$. Each class has one new element for each label, and each class *meets* 2α of the copies, two in every block:

	in a block of type a	in a block of type b
class k meets the copies at positions	$k-1,\ k+1$	$k+3,\ k+4$
class $\bar{k}$ meets the copies at positions	$k+2,\ k+3$	$k-1,\ k+1$

Read the other way: the copy at position i of a block of type a is met by the unbarred classes $i \pm 1$ and the barred classes $i+2, i+3$; of a block of type b, by the unbarred classes $i+1, i+2$ and the barred classes $i \pm 1$. So every copy is met by exactly two classes of each bar, and every class misses 3α copies (Figure 10).

In every copy it meets, the new element of a class with label t is incident with exactly one element of that copy: the point labelled t or the line labelled t. The thesis fixes which of the two by its figures, which draw each copy with an outer and an inner dot but do not say which dot stands for the points. One reading (*reading 2*, the one used in the lab's first computations) is that the unbarred classes sit on points and the barred classes on lines, in every copy. In incidence terms, Gómez's structure in reading 2 is:

- the points and lines of all 5α copies, with their incidences;
- for every point p of Γ and every $k \in \mathbb{Z}_5$, a new line $p^{(k)}$ through the copies p_c of p in the 2α copies Γ_c met by k;
- for every line L of Γ and every $k \in \mathbb{Z}_5$, a new point $L^{(\bar{k})}$ on the copies L_c of L in the 2α copies met by $\bar{k}$.

There are no other incidences; in particular no two new elements are incident. In the other reading of the figures (*reading 1*, which the conventions of the thesis's other figures suggest: the outer dot is the points in every copy) a new element of label t is incident with the line labelled t in the copies of one block type and with the point labelled t in the copies of the other, so the copies of one block type become copies of the dual Γ^D, and the pairing of point labels with line labels, which the thesis leaves open, now matters.

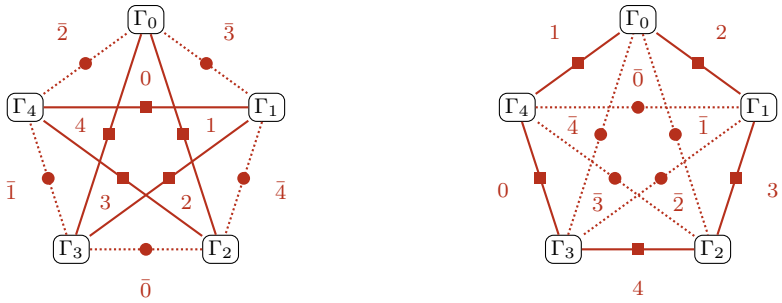
Counts. Every point and line of every copy gains two incidences (it carries one label, and exactly two classes that sit on its kind meet its copy), so it has degree $q+3$. A new element has degree $2\alpha \le q+3$. There are $5\alpha \cdot 2N$ old and $10N$ new elements, $5(\alpha+1)\cdot 2N$ in all:

cell	q	α	copies	order $5(\alpha+1)\cdot 2N$
$(10,8)$	7	5	25	1,176,480
$(11,8)$	8	5	25	2,246,940
$(12,8)$	9	6	30	4,650,100
$(14,8)$ and $(15,8)$	11	7	35	14,172,480
$(16,8)$	13	8	40	36,201,060

6.2. **What happens: the diameter is at least 9.** A new element is incident only with copies of *one* element of Γ. So passing through it lets a path change copy, but not move in Γ. This is the whole difficulty.

Theorem 6.2 (Theorem 1 and Corollary 3.1)**.** *In reading 2, for every new element x with label t (the point p for $p^{(k)}$, the line L for $L^{(\bar{k})}$), every copy $\Gamma_{c'}$ that x misses and every element v of Γ opposite t, the distance from x to $v_{c'}$ is at least 9. So the structure has diameter at least 9, for every q and every choice of block types.*

Proof. Send every element y_c of a copy to y, and every new element to its label. Along a path starting at x, an incidence inside a copy is sent to an incidence of Γ, and a passage copy, new



block of type a block of type b

FIGURE 10. Gómez's wiring of one block of five copies (after his Figures 2.21a,b; copy Γ_i is the copy at position i of the block). Each red segment is a class: its new elements meet the two copies at its ends. Squares on solid segments are the unbarred classes k (in reading 2, new lines through the copies of a point), circles on dotted segments the barred classes $\bar{k}$ (new points on the copies of a line). In a block of type a the unbarred classes are the diagonals and the barred classes the sides of the pentagon; in a block of type b the other way round. A new element meets the two copies at the ends of its class in every block, 2α copies in all.

element, copy uses two incidences and is sent to a single element (the new element is incident only with copies of its label). A path from x to $v_{c'}$ starts with an incidence from x to a copy of t in a copy that x meets; to reach $\Gamma_{c'}$ it must pass through at least one more new element. Removing the passages, its image is a walk in Γ from t to v, of length at least 6. So the path has length at least $1+6+2=9$ (Figure 11). □

The argument uses only that no two new elements are incident, that every new element meets copies of a single element, and that it misses some copy; it holds for any bipartite graph in place of the hexagon (with D_1+3 in place of 9), and it is formalised in Lean (`OMEGA_NOTE.md`, §3). In reading 1 with both block types a new element meets the point labelled t in some copies and the line labelled t in others, so its image may jump between the two; a count of the possible images still leaves elements far away:

Theorem 6.3 (Theorem 2)**.** *Let $q \geq 7$, and take the labelling of the thesis. In every reading, for every choice of block types and every pairing of point labels with line labels, every new element has, in every copy it misses, at least $q^5-5q^4-8q^3-12q^2-16q-10>0$ elements at distance at least 9.*

Why. Let x have label t, and let z be the element labelled t of the kind of the elements of $\Gamma_{c'}$ that x must reach within 7. A path of length at most 7 from x into $\Gamma_{c'}$ passes through 1, 2 or 3 new elements besides x and uses at most 4 incidences inside copies; each passage keeps the image or switches it between the point and the line with the same label. With no switch the image stays within distance 4 of z; a path that switches once starts at the other element with label t and reaches at most A elements opposite z; one that switches twice reaches at most B; with three switches it ends at z. Here, by Lemma 6.1(a), $A=\sum_{e_1+e_2\in\{0,2,4\}} s_{e_1}s_{e_2} = 5q^4+8q^3+6q^2+4q+2$ and $B=\sum_{e_1+e_2+e_3\leq 2} s_{e_1}s_{e_2}s_{e_3} = 6q^2+12q+7$, with $s_0=1$, $s_i=(q+1)q^{i-1}$. So at most $A+B+1$ of the q^5 elements opposite z are within 7, and for $q\geq 7$ this is less than q^5 (for example $15{,}459 < 16{,}807$ at $q=7$).

By computer, the diameter is exactly 9 for $q=2,3,5,7,9$ with both block types (for the patterns and readings computed), the eccentricity of a new element is 9 at the orders of all six table entries, and with a single block type the diameter is 10 ($q=2,3,5,9$; at $q=7$ some element has eccentricity 10) (`OMEGA_NOTE.md`, Computer-checked C2 and C3; checked again here for $q=2,3$).

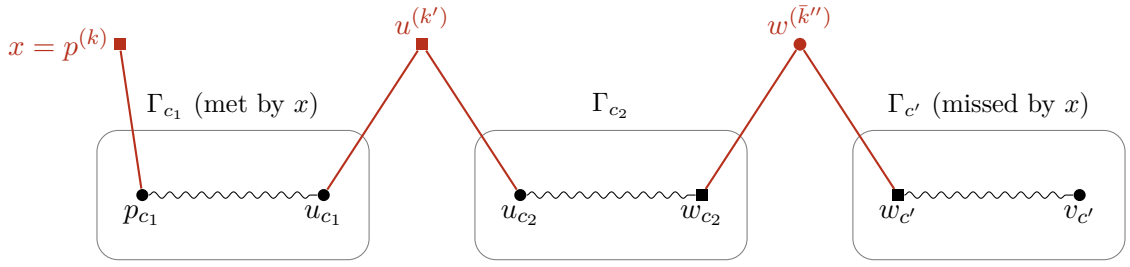


FIGURE 11. The projection (Theorem 6.2, reading 2). Wavy: walks inside one copy; red: new elements. A new element meets only the copies of its label, so a passage copy, new element, copy costs two incidences and does not move the image in Γ (u_{c_1} and u_{c_2} are copies of the same point). If v is opposite p and x misses $\Gamma_{c'}$, every path from x to $v_{c'}$ has length at least $1 + 6 + 2 = 9$.

6.3. **The repair: a labelling for each copy.** The thesis uses one labelling for all copies. The obvious repair keeps the wiring and gives every copy c its own labelling: bijections P_c and L_c from the labels to the points and lines of Γ, with the new element of label t incident, in copy c, with $P_c(t)$ or $L_c(t)$. Degrees and counts do not change. A passage from copy c to copy d through a new element of class Y now moves the image in Γ by the *transfer* $P_d P_c^{-1}$ (or $L_d L_c^{-1}$, $L_d P_c^{-1}$, $P_d L_c^{-1}$, according to what Y sits on in c and d), and Theorem 6.2 no longer applies. What is known:

- *At $q = 2$ and $q = 3$ the repair works* (`OMEGA_NOTE.md`, Computer-checked C7). Labellings found by annealing in the lab's table audit give diameter 8: at $q = 2$, block types ab, 1,890 elements, old elements of degree 5 and new ones of degree 4; at $q = 3$, block types aba, reading 1, 14,560 elements, 6-regular. We rebuilt the $q = 3$ structure from the audit's files and checked its diameter (Appendix A). These are far below the table at $(5, 8)$ and $(6, 8)$; they only show that a suitable labelling can reach $D_1 + 2$. We have no description of these labellings.
- *Relabelling a copy by an automorphism of Γ changes nothing* (`OMEGA_NOTE.md`, Lemma 5.2): if P_c and L_c are P_0 and L_0 followed by one collineation g_c of Γ, then $y_c \mapsto (g_c^{-1}(y))_c$ is an isomorphism onto Gómez's structure; the same holds with a duality g_c, if $P_c = g_c L_0$, $L_c = g_c P_0$ and the roles of points and lines in copy c are exchanged.
- *Geometric labellings cannot work for $q \geq 7$* (`OMEGA_NOTE.md`, Theorems 3 and 4). Call a labelling *geometric* if every transfer is induced by an automorphism of Γ: a collineation, or a duality, which exists for the split Cayley hexagon $H(q)$ only when q is a power of 3 [6]. For $q \geq 7$ the diameter is at least 9 for every geometric labelling, every reading, every choice of which kind each class sits on, and every choice of block types.
- *Open*: labellings that are not geometric, for every $q \geq 4$ (searches at $q = 4, 5$ found none of diameter 8); geometric labellings at $q = 4, 5$ (only computations); labellings chosen separately for each pair (class, copy); other wirings.

Why geometric labellings fail (combinatorially, for $q \geq 8$). After relabelling each copy by an automorphism (which changes nothing), the transfers of one bar, the *point bar*, are the identity and those of the other, the *line bar*, are collineations (`OMEGA_NOTE.md`, Lemma 5.3; for q not a power of 3 every class sits on one kind in all copies it meets). Collineations preserve distances. A path of length at most 7 from a new element x of the point bar with label t into a copy $\Gamma_{c'}$ it misses passes through one, two or three new elements; a passage through the point bar keeps the image, a passage through the line bar applies a collineation λ that depends only on the two copies. So every point of $\Gamma_{c'}$ within distance 7 of x is within distance 4 of t, or within distance 4 of $\lambda(t)$ for the collineation

λ of some copy c_1 met by x that shares a class of the line bar with c' (one passage through the line bar), or within distance 2 of one of at most m_2 further images of t (two passages; `OMEGA_NOTE.md`, Lemma 5.4). Call (x, c') *starved* if only α of the 2α copies met by x share a class of the other bar with c'; for every choice of block types starved pairs exist (`OMEGA_NOTE.md`, Lemma 5.6, a property of the wiring alone). By Lemma 6.1(b) these neighbourhoods then contain at most $\alpha q^4 + m_2 q^2$ of the q^5 elements opposite t, and a count of the two-passage words gives $m_2 \le 8\alpha^2 - 1$ in general and $m_2 \le 140$, 201 for $\alpha = 5, 6$. So for $q \ge 8$ every new element of a starved pair has at least $q^5 - \alpha q^4 - m_2 q^2$ elements at distance at least 9 in the missed copy: at least 3,328, 3,402, 11,253, 56,446 for $q = 8, 9, 11, 13$ (by hand for $q \ge 10$; for $q = 8, 9$ the bounds on m_2 are counts done by computer, which we repeated). At $q = 7$ the count is negative ($16{,}807 - 5 \cdot 2{,}401 - 140 \cdot 49 < 0$), and the proof uses the embedding of $H(7)$ in the quadric $Q(6, 7)$: the overlaps of the balls, and a character-sum bound on how many points a collineation without fixed points moves to points perpendicular to them (`OMEGA_NOTE.md`, §5.5). We do not reproduce it here.

6.4. **What can be claimed.** The six entries $(10, 8)$, $(11, 8)$, $(12, 8)$, $(14, 8)$, $(15, 8)$, $(16, 8)$ are the orders of Gómez's structures ΩH_q, $q = 7, 8, 9, 11, 11, 13$. As defined in the thesis (every copy of the hexagon carries the same labels, and a new element with label t is incident only with elements labelled t), they have diameter at least 9 for each of these q, under every reading of the figures and every pairing of point labels with line labels (Theorems 6.2 and 6.3, and computation). The paper the table cites was never published, as far as we could find, so these cells have no checkable structure of diameter 8. Giving each copy its own labelling repairs the construction at $q = 2, 3$; for $q \ge 7$ no geometric labelling does. (For $q = 7, 8, 11, 13$, reading 1 with both block types has no geometric labelling at all, since a transfer from points to lines would be a duality; there Theorem 6.3 covers the thesis labelling.) We do not claim that no structure with these parameters exists, nor that no labelling works.

7. Diameters 9 to 14: Haar structures

Beyond diameter 6 there is no generalized polygon with as many points as lines to start from. By the theorem of Feit and Higman [14], a generalized n-gon with $s, t \ge 2$ has $n \in \{3, 4, 6, 8\}$, and for $n = 8$ the orders s and t differ ($2st$ is a square). Sections 4–6 built diameters 5, 7 and 8 from several copies of a quadrangle or a hexagon. Our values at diameters 9 to 14 come from somewhere else: from incidence structures with a group of automorphisms acting regularly on the points and on the lines. Such a structure is determined by a finite group G and a set S of Δ elements of G. Nothing is deleted or added; the work is in choosing G and S, and this is done by computer search. So, unlike Sections 1–6, this section has no theorem that produces a family: each value is one structure, found by a search and checked by two programs. We name the groups, but every argument uses only the group law. The sources are `research/frontier/SIZE_LAW.md` (the size law), `research/structured/STRUCTURE.md` (the orbit search), `research/theory/` (what is proved), `research/frontier/beyond11/ODD_D_HOLES.md` and `FROBENIUS.md` (branch `claude/beyond-d11-d12-new-cells-3t3djx`), `research/frontier/beyond/BEYOND_TABLE.md` (branches `claude/beyond-d11-d12-new-cells-3t3djx` and `claude/beyond-d13-search-z35z61`), the results table of `README.md` and the ledger `research/RESULTS.md` on `master`.

7.1. Haar structures.

Definition 7.1 (Haar structure)**.** Let G be a finite group with identity e, and $S \subseteq G$ a set of Δ elements. The *Haar structure* $H(G, S)$ has a point p_g and a line L_g for every $g \in G$, and

$$p_g \,\mathrm{I}\, L_h \iff hg^{-1} \in S.$$

So the lines through p_g are the Δ lines L_{sg}, $s \in S$, and the points of L_h are the Δ points $p_{s^{-1}h}$, $s \in S$; every element has degree Δ, and there are $2|G|$ elements. The incidence graph is the

Haar graph of G and S [15]: in the lab's files its vertices are $(g, A) = p_g$ and $(h, B) = L_h$, with $(g, A) \sim (sg, B)$.

Lemma 7.2. (a) *For $x \in G$, the map $\rho_x \colon p_g \mapsto p_{gx}$, $L_h \mapsto L_{hx}$ is an automorphism of $H(G, S)$. The group of these maps acts regularly on the points and regularly on the lines.*
(b) *Conversely, if a group G of automorphisms of an incidence structure maps points to points and lines to lines and acts regularly on both, the structure is a Haar structure of G.*
(c) *$H(G, S)$ is isomorphic to $H(G, Ss_0^{-1})$ for every $s_0 \in S$ (by $p_g \mapsto p_{s_0 g}$, $L_h \mapsto L_h$), and to $H(G, \varphi(S))$ for every automorphism φ of G (by $p_g \mapsto p_{\varphi(g)}$, $L_h \mapsto L_{\varphi(h)}$).*

Proof. (a) $(hx)(gx)^{-1} = hg^{-1}$, and ρ_x maps p_e to p_x. (b) Write the action on the right, fix a point p and a line L, and name the points p^g and the lines L^h $(g, h \in G)$. Then p^g I L^h iff $p^{gh^{-1}}$ I L, that is, iff $hg^{-1} \in S$ with $S = \{t^{-1} : p^t \text{ I } L\}$. (c) $hg^{-1} \in S$ iff $h(s_0 g)^{-1} \in Ss_0^{-1}$, and $\varphi(h)\varphi(g)^{-1} = \varphi(hg^{-1})$. □

By (c) we may and do assume $e \in S$; all the lab's files are written that way. A first example, at diameter 3: if G is the cyclic group $\mathbb{Z}_{q^2+q+1}$ and S is a perfect difference set (every nonzero element of G is a difference of two elements of S in exactly one way), then two points $p_g, p_{g'}$ have exactly one common line, two lines exactly one common point, and $H(G, S)$ is a projective plane of order q (Singer [16]). So the Haar structures include the Desarguesian planes of Section 1; there the group was invisible.

7.2. **Distances, and why two searches are enough.** A path from p_e alternates points and lines: p_e, L_{s_1}, $p_{s_2^{-1}s_1}$, $L_{s_3 s_2^{-1} s_1}$, and so on. So the elements reached are products of elements of S and their inverses, with alternating signs. For sets $X, Y \subseteq G$ write $XY = \{xy : x \in X,\, y \in Y\}$ and $X^{-1} = \{x^{-1} : x \in X\}$.

Lemma 7.3 (Two searches)**.** *Let $e \in S$, and let $k \geq 0$.*
(a) *$d(p_e, p_g) \leq 2k$ iff $g \in (S^{-1}S)^k$; $d(L_e, L_h) \leq 2k$ iff $h \in (SS^{-1})^k$; $d(p_e, L_h) \leq 2k+1$ iff $h \in S(S^{-1}S)^k$.*
(b) *Every element is within $\max(\varepsilon(p_e), \varepsilon(L_e))$ of every other, where ε is the eccentricity; so $\operatorname{diam} H(G, S) = \max(\varepsilon(p_e), \varepsilon(L_e))$.*
(c) *$\operatorname{diam} H(G, S) \leq 2k$ iff $S(S^{-1}S)^{k-1} = G$ ($k \geq 1$): one condition, every line within $2k-1$ of p_e.*
(d) *$\operatorname{diam} H(G, S) \leq 2k+1$ iff $(S^{-1}S)^k = G$ and $(SS^{-1})^k = G$: two conditions, every point within $2k$ of p_e and every line within $2k$ of L_e.*

Proof. (a) By induction on the path, as above; since $e \in S^{-1}S$ and $e \in SS^{-1}$, the sets grow with k, so "at most $2k$" and "a product of exactly k factors" agree. (b) $\rho_{g^{-1}}$ maps p_g to p_e and $\rho_{h^{-1}}$ maps L_h to L_e, and automorphisms keep distances. (c) As in the proof of Lemma 0.1: point–line distances are odd, and two points (or two lines) are within $2k$ as soon as every point–line pair is within $2k-1$. By (a) applied after $\rho_{g^{-1}}$, $d(p_g, L_h) = d(p_e, L_{hg^{-1}})$, so all point–line pairs are within $2k-1$ iff every line is within $2k-1$ of p_e. (d) As in Lemma 5.1: the diameter is at most $2k+1$ iff any two elements of the same kind are within $2k$; by (b) it suffices to look from p_e and from L_e, and (a) translates this. □

So one breadth-first search from p_e decides an even diameter, and two, from p_e and from L_e, decide an odd one. Each search visits every element once, instead of once for each of the $2|G|$ starting elements: this is what makes structures with $3 \cdot 10^{10}$ elements checkable (§7.7). At odd D both searches are needed: for $G = \mathbb{Z}_{13} \rtimes_3 \mathbb{Z}_3$ (elements (a, j), $(a, j)(b, l) = (a + 3^j b,\, j + l)$) and $S = \{(0,0), (0,1), (1,0)\}$, $(SS^{-1})^3 = G$ but $(S^{-1}S)^3$ misses two elements, so $\varepsilon(L_e) = 7$ and $\varepsilon(p_e) = 8$ (`check_haar.py`). The "if" directions of (c) and (d) are proved in Lean for every k (`lean/BipartiteDD/Haar/GraphD.lean`, `haar_ediam_le_even` and `haar_ediam_le_odd`).

The counting bound. A product $s_\ell^{\pm 1}\cdots s_2^{-1}s_1$ with $s_{i+1}\neq s_i$ is a path that never steps straight back, and there are $\Delta(\Delta-1)^{\ell-1}$ of length $\ell\geq 1$. Every element of G must be such a product of one of the lengths in (c) or (d), that is, of length at most $D-1$ and of the parity of $D-1$. So $|G|\leq W(\Delta,D)$, where

$$W(\Delta,2k)=\Delta\sum_{i=0}^{k-1}(\Delta-1)^{2i},\qquad W(\Delta,2k+1)=1+\sum_{i=1}^{k}\Delta(\Delta-1)^{2i-1}.$$

Expanding $\Delta=(\Delta-1)+1$ gives $W(\Delta,D)=\sum_{i=0}^{D-1}(\Delta-1)^i=M_b(\Delta,D)/2$ in both cases, where M_b is the bipartite Moore bound. So for Haar structures the counting bound is the Moore bound itself, and the Singer planes attain it at $D=3$. The longest words, of length $D-1$, are about $\Delta(\Delta-1)^{D-2}$ in number and carry almost all of W.

7.3. **Why the orders are what they are: the size law.** `SIZE_LAW.md` found that the best Haar structures found by search follow $|G|\approx C_D(\Delta-1)^{D-2}$ within each diameter, within a few per cent, with a C_D that rises slowly with Δ and falls with D. The explanation is a coupon-collector count. Write $N=|G|$, $W=W(\Delta,D)$, $\lambda=W/N$, and pretend that the W words of the last condition are W independent uniform elements of G. Then the expected number of elements of G that no word reaches is

$$\mu=N(1-1/N)^W\approx Ne^{-\lambda},$$

and the number missed is roughly Poisson with mean μ: one random S works with probability about $e^{-\mu}$. A search that tries many sets S can absorb a small mean μ^*, which depends on how long it runs. Solving $Ne^{-W/N}=\mu^*$ for N, and using $W=M_b/2$,

$$2N=\frac{M_b(\Delta,D)}{\ln(N/\mu^*)}.$$

With $\mu:=Ne^{-W/N}$ taken as a definition this is an identity; the content is that at our values μ stays between about 3 and 61 (Appendix B; four values stopped at a size cap have $\mu\approx 0$), while $\ln N$ grows from about 10 to 23. So a Haar structure found by search has a fraction about $1/\ln(N/\mu)$ of the Moore bound, falling as N grows: from 17% ($\Delta=5$) to 5% ($\Delta=20$) at $D=9$, and from 10% ($\Delta=4$) to 5% ($\Delta=8$) at $D=14$. Fitted on the best structure of each cell then available (19 cells at $D=5,8,9,10$), $\ln\mu^*=4.34-0.123\ln N$; with this fit `SIZE_LAW.md` predicted, before running them, the orders reached in seven cells not tried before, within 1 to 3%, and "a run solves iff $\mu\leq\mu^*$" held for every run of that test (§3.2 there).

This is a heuristic, not a theorem. For D fixed and Δ large, a threshold theorem of this kind is proved for every finite group (`research/theory/THRESHOLD.md`, Theorems 3 and 4; we have not checked these proofs here). In the regime of the records, Δ fixed and D growing, it is open, as is any proof that $\operatorname{diam}H(G,S)$ is close to $\log_{\Delta-1}|G|$ for random S (`research/theory/SUMMARY.md`). The model also says where the constant comes from: the elements that are missed are not uniform, and the group decides which ones are at risk.

Lemma 7.4 (Odd diameter: inverses and central elements)**.** *Let $e\in S$, $h\geq 1$, $X_A=(S^{-1}S)^h$ and $X_B=(SS^{-1})^h$.*

(a) *$X_A=X_A^{-1}$ and $X_B=X_B^{-1}$. The words of length $2h$ reaching x (sequences $s_1,\dots,s_{2h}$ with $s_{2h}^{-1}s_{2h-1}\cdots s_2^{-1}s_1=x$, as in Lemma 7.3) are in bijection with those reaching x^{-1}, by reversing the sequence; for an involution x the reversal permutes the words reaching x, and has no fixed sequence.*

(b) *A central element c lies in X_A iff it lies in X_B.*

Proof. (a) The inverse of $s_{2h}^{-1}s_{2h-1}\cdots s_2^{-1}s_1$ is $s_1^{-1}s_2\cdots s_{2h-1}^{-1}s_{2h}$, the word of the reversed sequence. A sequence equal to its reverse gives e, since the middle letters cancel and then the next ones, and

so on; e is not an involution. The same for X_B. (b) If $c = s_{2h}^{-1}(s_{2h-1}\cdots s_1)$, conjugating by s_{2h}^{-1} gives $c = (s_{2h-1}\cdots s_1)s_{2h}^{-1}$, a product of h factors of SS^{-1}; the converse is the same. □

So at odd D the words reaching an involution come in pairs, and in the Poisson picture an involution is missed with probability about $e^{-\lambda/2}$ instead of $e^{-\lambda}$; a central element is missed on both sides at once, and the words reaching it are also closed under moving two letters from the front to the back, so it is missed with probability about $e^{-\lambda/h}$ (`ODD_D_HOLES.md`, where these rates are measured). $\mathrm{PSL}(2,p)$ has about $p^2/2$ involutions and $\mathrm{SL}(2,p)$ one, $-I$. At $(11,11)$, three random symmetric S in $\mathrm{PSL}(2,1063)$ ($\mu = 5.5$) missed 117 to 159 involutions, and three in $\mathrm{SL}(2,251)\times\mathbb{Z}_{37}$ ($\mu = 3.3$) missed 2 to 22 elements in all. This is why the $D = 11$ values for $\Delta = 7,\dots,15$ use $\mathrm{SL}(2,p)\times\mathbb{Z}_n$, and why no odd-$D$ value uses $\mathrm{PSL}(2,p)$ alone; metacyclic groups of odd order have no involutions and a trivial centre. At even D both effects disappear, since the words of Lemma 7.3(c) have odd length and reversing one changes its side ($\mathrm{PSL}(2,389)$ alone solved $(10,10)$ in 385 evaluations, `STRUCTURE.md`, §2).

7.4. **Orbit search: S closed under an automorphism.** Let α be an automorphism of G of prime order k, and take

$$S = F \cup O_1 \cup \cdots \cup O_m, \qquad F \subseteq \mathrm{Fix}(\alpha),\ e \in F,\ |F| = f, \qquad O_j = \{x_j, \alpha(x_j), \dots, \alpha^{k-1}(x_j)\},$$

with x_j not fixed by α, so that $\Delta = f + km$. (When $k \mid \Delta$ the lab also uses $f = 0$: S is Δ/k orbits and does not contain e; by Lemma 7.2(c) the files then store Ss_0^{-1}.)

Lemma 7.5. *If $\alpha(S) = S$, then $p_g \mapsto p_{\alpha(g)}$, $L_h \mapsto L_{\alpha(h)}$ is an automorphism of $H(G,S)$ fixing p_e and L_e. So for every r the set of elements at distance more than r from p_e (or from L_e) is a union of α-orbits.*

Proof. $\alpha(h)\alpha(g)^{-1} = \alpha(hg^{-1})$, and $\alpha(S) = S$. □

So the elements a symmetric S misses come in whole α-orbits: k at a time, or one fixed element at a time. In the Poisson picture the number of missed orbits has mean about μ/k, plus about $e^{-\lambda/k}$ for each fixed element (the k words that α permutes all reach the same fixed element); a random symmetric S works with probability about $e^{-\mu/k}$ instead of $e^{-\mu}$, and the search moves the m representatives x_j instead of $\Delta - 1$ free elements. This does not raise the counting ceiling: there are only about N^m symmetric sets, against $N^{\Delta-1}$ in all, and the first-moment estimate gives the same ceiling for both. It makes the band below the ceiling searchable. On the same groups and seeds at $(7,8)$, orbit search solved $\mu = 34$ in 1.7–$5.8\cdot10^4$ evaluations of S, where random S had not succeeded after 1.2–$1.6\cdot10^6$; the reachable μ roughly doubled, which is 5 to 11% in N (`STRUCTURE.md`, §§0, 3).

What is used (`STRUCTURE.md`, §2, and the beyond jobs): $G = Q\times\mathbb{Z}_n$ with $Q = \mathrm{SL}(2,p)$ or $\mathrm{PSL}(2,p)$ (p prime; also $\mathrm{PSL}(2,81)$, $\mathrm{PSL}(2,243)$ and products such as $\mathrm{PSL}(2,7)\times\mathrm{SL}(2,7)$), and $k = 3$, with α conjugation by an element of order 3 (modulo the centre) on Q times $z \mapsto rz$ on $\mathbb{Z}_n$, r of order 3 modulo n; then $f = 1$ or 2 when $\Delta \equiv 1$ or 2 modulo 3, and $f = 0$ (or $f = 3$, at $\Delta = 18$) when $3 \mid \Delta$; the fixed generators other than e lie in the torus that centralizes the conjugating element. Five cells at $D = 9, 10$ use $k = 5$: $(10,9)$, $(10,10)$, $(15,10)$ and $(16,10)$ with conjugations, and $(11,10)$, $\mathrm{PSL}(2,243)\times\mathbb{Z}_{11}$, with the field automorphism $x \mapsto x^3$ of $\mathrm{GF}(243)$ applied to the matrix entries in place of a conjugation. Three rules came out of the experiments: k must be odd (for an involution α, two different words collide whenever $\alpha(u) = u^{-1}$ for a product u of generators, and this happens for a fraction about $1/p$ of all u); α must move every small quotient of G (an orbit has a single image in a quotient on which α acts trivially), hence every prime factor of n is $\equiv 1$ modulo k (true for all our values); and α should have few fixed points (a conjugation fixes a torus, about p elements). For the nine values of Table 5 with a symmetric S containing e that it tests, from $(7,9)$ to $(14,12)$ (seven with $k = 3$, two with $k = 5$), `check_haar.py` recovers α from S alone, by linear algebra, and checks $\alpha(S) = S$.

7.5. **Small degrees: metacyclic groups and lifts of cubic configurations.** At $\Delta = 4, 5$ (and $\Delta = 6$ at $D = 11$) the best groups are metacyclic, $\mathbb{Z}_m \rtimes_r \mathbb{Z}_k$ with $(a, j)(b, l) = (a + r^j b,\, j + l)$ and $r^k \equiv 1 \bmod m$, mostly of odd order and mostly Frobenius groups (m prime, r of order k), with random S and no symmetry. A Frobenius group of odd order has no involutions and a trivial centre, so Lemma 7.4 costs nothing, and at these small orders it mixes like a random group, while $\mathrm{SL}(2, p)$ for small p has many short relations (`FROBENIUS.md`). At $(4, 11)$ the metacyclic value sits at $\mu = 44$, where SL and PSL groups stall at $\mu \approx 8$–15.

At $\Delta = 3$ the dipole (one point, one line and three incidences between them, whose lifts are the Haar structures) is not the best base. Let B be a small incidence structure with all degrees 3, and give each flag (P, L) of B an element $v(P, L)$ of a group Γ. The *lift* has points (P, γ) and lines (L, γ), $\gamma \in \Gamma$, with (P, γ) I $(L, \gamma v(P, L))$ for every flag (P, L) of B; all degrees are again 3, and Γ acts on it with $|B|$ orbits. The Haar structure $H(G, S)$ is the lift of the dipole with the elements of S on its flags. Our values: $(3, 13)$, the Desargues configuration (10 points, 10 lines; incidence graph the generalized Petersen graph $GP(10, 3)$) lifted by $\mathbb{Z}_{170}$; $(3, 14)$, the Pappus configuration (9 points, 9 lines) lifted by $\mathbb{Z}_{317}$; $(3, 12)$, the structure whose 4 lines are the four 3-subsets of 4 points, lifted by $\mathbb{Z}_{37} \rtimes_6 \mathbb{Z}_8$; $(3, 10)$, a base with 5 points and 5 lines lifted by $\mathbb{Z}_3 \times \mathrm{He}_3$ (He_3 the Heisenberg group of order 27); and $(3, 11)$ is the Haar structure of $\mathbb{Z}_{67} \rtimes_9 \mathbb{Z}_{11}$. The voltages were found by annealing, with an exhaustive one-change search at the end for $(3, 13)$, $(3, 14)$.

7.6. **Our values at diameters 9 to 14.** Table 5 gives, for every cell with $9 \le D \le 14$ and $3 \le \Delta \le 20$ in which the lab has a structure, the number of elements of the largest one; Appendix B gives for each the group, the shape of S, μ and the checks. The values are those of the results table of `README.md` on `master` ($D = 9, 10$, $\Delta \le 16$), of the ledger `research/RESULTS.md`, which is ahead of the README in three cells ($(5, 9)$, $(7, 10)$, $(8, 10)$: 29,756, 2,298,240, 8,043,840 against 29,540, 2,237,760, 7,818,096), and of `BEYOND_TABLE.md` on the beyond branches ($\Delta = 17, \ldots, 20$ at $D = 9, 10$; $D = 11, \ldots, 14$). Where these disagree we took the largest value with a check log: $(4, 12)$ 74,066 from `claude/beyond-d13-search-z35z61` (the ledger has 70,176), $(3, 11)$ 1,474 from `BEYOND_TABLE.md` (the ledger has 1,344), and $(3, 12)$ 2,368 from the ledger (`BEYOND_TABLE.md` has 2,176). At $(14, 12)$ the structure found for $\Delta = 14$ has 29,996,171,040 elements, fewer than the $(13, 12)$ structure, whose degree 13 is at most 14; so the value of that cell is the $(13, 12)$ one.

The 2013 table [12] stops at $D = 10$ and $\Delta = 16$. In its cells our values are larger in each case: by 6.5 to 32.9% at $D = 9$ ($\Delta = 5, \ldots, 16$) and by 1.3 to 27.7% at $D = 10$ ($\Delta = 3, \ldots, 16$). For the other cells we know of no published bipartite value (literature card r2/d10 of the job `new-cells-beyond`), so they are first values, with one exception: the Foster census of cubic edge-transitive graphs lists a bipartite graph CSS(6272, 7) with 6,272 vertices and diameter 14, larger than our 5,706 at $(3, 14)$ (`research/squads/own-small/newcells/README.md`; we have not checked it). The census graphs listed there for $D = 11, 12, 13$ (1,250, 2,304, 3,360 vertices) are smaller than ours. Four values ($\circ$) are the first S that solved at a size limit of the job (20 or 30 billion elements), with $\mu \approx 0$; they say nothing about how far a search could go. The values at $\Delta = 17, \ldots, 20$, $D = 9, 10$ were taken at $\mu \approx 3$ and not pushed. In the other cells μ lies between about 5 and 61.

7.7. **How the values were checked.** Every Haar value was checked by programs that rebuild the structure from its solution file (the group from its definition, the elements of S from their descriptions) and run the searches of Lemma 7.3 from p_e and from L_e; the lifts and the smallest Haar structures were also checked by a breadth-first search from every element, which uses no symmetry. Each is a separate program from the search that found the structure and builds the group itself; what each file's header says about its independence is noted. They are (the letters are those of Appendix B):

- B `research/frontier/build_haar2.py`: rebuilds the whole incidence graph with numpy, for every group family the searches use; it fits in memory up to about $6 \cdot 10^7$ elements.

Δ	$D=9$	$D=10$	$D=11$	$D=12$	$D=13$	$D=14$
3	–	810†	1,474$^{\ddagger *}$	2,368†	3,400$^{\dagger *}$	5,706$^{\dagger *}$
4	–	12,416‡	30,334‡	74,066$^{\ddagger *}$	192,382$^{\ddagger *}$	497,042$^{\ddagger *}$
5	29,756‡	95,472‡	311,426‡	1,058,342‡	3,594,910$^{\ddagger *}$	12,795,166$^{\ddagger *}$
6	127,296	524,160	2,122,518‡	9,366,240	41,734,392*	187,990,656*
7	462,840	2,298,240	11,012,976	59,838,504	312,227,760*	1,674,490,944*
8	1,341,600	8,043,840	45,329,808	284,095,056	1,739,514,000*	11,325,238,056*
9	3,499,776	24,099,600	155,250,816	1,105,785,120	7,924,036,032*	–
10	8,418,120	64,294,560	460,691,568	3,712,424,352	19,980,294,768$^{\circ *}$	–
11	17,323,152	157,835,304	1,241,299,248	11,189,148,096	–	–
12	35,445,648	332,797,296	2,976,066,240	29,996,171,040	–	–
13	67,742,688	683,943,456	6,781,213,392	29,997,644,040°	–	–
14	120,043,728	1,350,537,864	14,381,398,080	29,997,644,040$^{\S \circ}$	–	–
15	208,483,200	2,566,531,440	28,655,908,512	–	–	–
16	348,207,600	4,708,566,360	29,961,590,400$^{\circ *}$	–	–	–
17	501,707,400	7,022,052,336	–	–	–	–
18	790,626,816	11,787,972,672	–	–	–	–
19	1,218,394,320	19,233,722,496	–	–	–	–
20	1,823,175,864	30,518,778,360	–	–	–	–

TABLE 5. Our largest structures with maximum degree Δ and diameter D, $9 \le D \le 14$ (number of elements, that is, of vertices of the incidence graph). Unmarked: Haar structures of $\mathrm{SL}(2,q)$, $\mathrm{PSL}(2,q)$ or products, times a cyclic group, with S closed under an automorphism of order 3 or 5; ‡: Haar structures of metacyclic groups (or a product of two), random S; †: lifts of cubic configurations, not Haar structures; ∘: the first S found at a size cap of the search (here $\mu \approx 0$); ∗: not yet in the lab's ledger (only on the beyond branches); §: the $(13,12)$ structure. $(3,9)$ and $(4,9)$: no value of ours above the 2013 table (506 and 4,970).

- I `research/frontier/check_haar_indep.py`: its own field and matrix arithmetic, one permutation of G per generator; SL, PSL, PGL over $\mathrm{GF}(q)$ and products.
- S `research/frontier/haarcheck_sl.c` and P `research/structured/tools/haarcheck_psl.c`: low-memory checkers for $\mathrm{SL}(2,p) \times \mathbb{Z}_n$ and $\mathrm{PSL}(2,p) \times \mathbb{Z}_n$, four bitsets of $|G|$ bits per search; P was modelled on S, so the two are never counted as a pair. T `research/frontier/haarcheck2.c`: another low-memory checker for $\mathrm{SL}(2,p) \times \mathbb{Z}_n$; its numbering of $\mathrm{SL}(2,p)$ uses the same formula as the search program `haar_gen3.c`, while S numbers the elements its own way.
- G `research/frontier/haarcheck_gen.c`: a low-memory checker for SL and PSL over $\mathrm{GF}(q)$ and products of two, times $\mathbb{Z}_n$, which tests the field and the group law at start-up; its header says its code is not shared with B, I or T.
- X `bigcheck_indep.c` (branch `claude/bigcheck-indep-haar-cdziu4`): written without reading the other checkers or the search programs.
- M `metacheck.py`, `metacheck2.py` (branch `claude/beyond-d13-search-z35z61`): metacyclic groups and products of two, with their own group code, written in the session that also wrote the search program `hmeta`.
- A `research/lifts/coordinator_check/apsp.c`: a search from every vertex of an edge list, no symmetry assumed; it reports the exact diameter.
- L the Lean proofs in `lean/BipartiteDD/Haar/`: kernel-checked certificates for the covering conditions of Lemma 7.3.
- H `research/synthetic/checks/check_haar.py`, written for this note from the definitions: it rebuilds the Haar structures with at most $2 \cdot 10^7$ group elements and runs the two searches, and runs a search from every vertex on the lifts and on $(3,11)$.

The logs are on `master`: the coordinator's harvest checks `research/frontier/runs/harvest_check_2026-10-*.txt` and `harvest_check_2026-11-01.txt`, `research/frontier/runs/cloudcheck/`, `research/coordinator_check/` and `research/lifts/coordinator_check/`; a check marked * has its log only on a beyond branch or a verification branch (the `BEYOND_TABLE.md` rows give the paths), and a check marked ′ was run for this note with the lab's checker, with its log in `research/synthetic/checks/haar/`. Every log names the file it checked, and the sha256 of the solution files agree.

In all, 45 values have two independent checkers with logs on `master`: the Haar values at $D = 9$ with $\Delta \geq 9$, at $D = 10$ with $\Delta \geq 7$, at $D = 11$ with $7 \leq \Delta \leq 15$ and at $D = 12$ with $6 \leq \Delta \leq 14$, and the lift at $(3, 12)$. Seven more, $(5, 9)$, $(6, 9)$, $(7, 9)$, $(3, 10)$, $(4, 10)$, $(5, 10)$ and $(6, 10)$, have one checker and a Lean proof. $(16, 11)$ has two (G, and P in two halves), with all logs on branches. The other 21 values had, before this note, one check, or a second one with its log only on a branch: $(8, 9)$, $(6, 11)$ and $(5, 12)$ only B (run twice); $(3, 11)$, $(3, 13)$ and $(3, 14)$ only searches from every vertex of one edge list; $(5, 13)$ and $(5, 14)$ only M; $(4, 11)$, $(5, 11)$, $(4, 12)$, $(4, 13)$ and $(4, 14)$ a second check on a branch; and $(\Delta, 13)$ for $\Delta = 6, \ldots, 10$ and $(\Delta, 14)$ for $\Delta = 6, 7, 8$ only G. For each of these a second independent check was run for this note, with its log on this note's branch: H for the first 13, and S or P for the last eight.

Appendix A. Coordinates and checks

Coordinates. Explicit coordinates for every construction: `research/theorems/POLISHED.md` (§2 for the hats, §§3–5 for diameter 4, §§6–7 for diameter 6; on branch `claude/thm-polish-2oddll`), `research/plus2/coords/coords.tex` (the hats, with a hand-checkable proof), `research/theorems/D7_HEXAGON.md` (diameter 7; branch `claude/d7-hexagon-theorems-4w1prt`), `research/theorems/D5_QUADRANGLE.md` (diameter 5: §1.2 for $W(q)$ and $Q(4, q)$, Lemma 1.1 for an explicit ovoid of $Q(4, q)$, Lemma 1.3 for the $3q - 1$ pairwise non-collinear points of $W(q)$, Lemma 1.4 for a duality of $W(q)$, q even; branch `claude/d5-relax-searches-osr72a`), and `research/theorems/OMEGA_NOTE.md` (diameter 8: §1.1 for $H(q)$ in the quadric $Q(6, q)$, §5.5 for the case $q = 7$; branch `claude/omega-d8-note-xrbqux`).

The completing matching of [1]. In coordinates, [1] (arXiv v1, p. 8) complete the three-hat structure by the pairs $(x, y) \sim [x, y]$ and $P_i \sim L_i$; with the index set $i \in \mathrm{GF}(q) \setminus \{0, 1, a\}$ printed there, the four elements P_1, P_a, L_1, L_a keep degree $q + 1$, and $i \in \mathrm{GF}(q)^*$ gives a regular structure (`POLISHED.md`, Remark 2.7; we have not seen the journal version). Neither the order nor the diameter depends on the matching (§1.5).

Checks. The scripts are in `research/synthetic/checks/`, each with its output in a `.out` file. They build the classical geometries from coordinates (`check_haar.py`: the groups from their definitions) and then use only incidence; each runs in a few seconds, except `check_d8.py` and `check_haar.py` (about three minutes each).

- `check_d3.py` (Section 1): both hat constructions from rules (R1)–(R6) on a plane given by its incidence relation; equality with the coordinate construction of `coords.py` for $q = 4, 5, 7, 8, 9$ and every admissible (a, c); the gadget tables; the matching of Figure 1; the crossed doubling and the patches (Proposition 1.2, Lemma 1.3, including the extra far pair at $q = 3$); the broken pairs and their repairs; every line of Table 4; degrees and the deficient elements (Lemma 1.4); the duality (Lemma 1.6); diameter 3; the changes of §1.7. Planes: $\mathrm{PG}(2, q)$, $q = 3, 4, 5, 7, 8, 9$, and the two planes of order 9 over the Dickson nearfield (lines $y = x \circ m + b$ and $y = m \circ x + b$), with random frames.
- `check_d4.py` (Section 2): Theorem 2.2 on random weights, shadow lines and stand-ins in $W(3)$ and AS(3); AS(q) built from $W(q)$ by Payne derivation using only incidence, and properties (E1), (E2) of the two quadrics, for $q = 3, 5, 7$; the 2-ovoid and spread of $W(q)$ and the explicit

completion, $q = 3, 5$; Delorme's construction on $W(3)$; ovoid and spread of $W(2)$; the $(7, 4)$ example.

- `check_d6.py` (Section 3): in $H(2)$ and $H(3)$: the spread, free points, whiskers, an exact whisker set (the formula of `POLISHED.md` for $q = 3$, an exact-cover search for $q = 2$), (L0) with equality, (L1), the components, the counting identity, the doubling (diameter 6, maximum degree $q + m$, the count), the sheets, and Theorem R. `check_d6_even.py`: the characteristic-2 whisker formula (`POLISHED.md`, Theorem 6.14) in $H(4)$ gives an exact whisker set, and the doubling with $m = 2$ has 3524 elements, maximum degree 6 and diameter 6.
- `check_d7.py` (Section 4): Theorem 4.2 with random matchings on $H(2)$, $r = 1, 2$; Theorem 4.5 on $H(3)$ (a spread's point set and a non-collinear set are repaired, a turning triple is not); Theorem 4.6 on $H(3)$ with a local duality at three pairwise opposite points (and a random bijection at the same set is not repaired); a three-copy structure with hubs on $H(3)$, diameter 7.
- `check_d5.py` (Section 5): $W(q)$ from the symplectic form of `D5_QUADRANGLE.md` and $Q(4, q)$ as its dual, $q = 2, 3, 5, 7$ (checked to be quadrangles for $q \leq 5$); Theorem 5.3 on random level structures ($q = 2, 3$, $r = 1, 2$); Theorem 5.5 against a direct distance computation on 416 transitions (among them transitions failing only (E2) and only (E3)); Theorem 5.6 for identity transitions ($q = 3$) and for the duality of $W(2)$; the two-point recipe, Theorem 5.7 and the remark on (H2) ($q = 3, 5$); triad centres in $W(q)$ and $Q(4, q)$ ($q = 3$ all triads, $q = 5$ a sample); Propositions 5.8 and 5.9 ($q = 3, 5$); the hub graphs (Y1), (Y2) ($s = 2, 3$); the slot vectors and the six family values; seven copies $HHHDHHD$ over $Q(4, 5)$ with a partner pair meeting at an ovoid point and hubs placed as in `D5_QUADRANGLE.md`, Lemma 7.3 (2,188 elements, diameter 5); and the family of §5.9 at $q = 7$ in full (4,044 elements, maximum degree 10, diameter 5).
- `check_d8.py` (Section 6): Lemma 6.1 on $H(2)$, $H(3)$; Gómez's wiring for all block types, both readings and random pairings ($q = 2, 3$): degrees, counts, colouring; the far sets of Theorem 6.2; exact diameters 9 ($q = 2$: ab; $q = 3$: aba; both readings) and 10 (one block type); the constants of Theorem 6.3; the starved pairs and the counts of two-passage words for $\alpha = 5, \ldots, 8$ and every choice of block types (largest m_2 of a starved pair 140, 201, 272, 353; in every choice some starved pair with $m_2 \leq 134$, 197, 262, 347), written from the wiring alone; `OMEGA_NOTE.md`, Lemma 5.2, at $q = 2$ with collineations $(v, a, w) \mapsto (Av, a, A^{-T}w)$; and the $q = 3$ labelling of the table audit (`research/projects/table-audit/omega/labs/q3_aba_LLL_G13.lab` on branch `claude/table-audit-2-p7x6hn`, with the audit's base hexagon, both copied into `checks/`), rebuilt from the definition: 14,560 elements, 6-regular, diameter 8.
- `check_haar.py` (Section 7): Lemmas 7.2 and 7.3 on random S in seven small groups ($\mathrm{SL}(2, 3)$, $\mathrm{SL}(2, 5) \times \mathbb{Z}_3$, $\mathrm{PSL}(2, 7)$, S_4, the dihedral group of order 12, $\mathbb{Z}_{31} \rtimes_5 \mathbb{Z}_3$, $\mathbb{Z}_7 \rtimes_2 \mathbb{Z}_9$): right multiplication, the diameter from all pairs against the two searches, the word conditions for $D = 2, \ldots, 14$; the example in $\mathbb{Z}_{13} \rtimes_3 \mathbb{Z}_3$ where the two odd-D conditions differ; the isomorphisms of Lemma 7.2(c); the Singer planes of orders 2 to 5; the word counts and $W = M_b/2$; Lemmas 7.4 and 7.5 in $\mathrm{SL}(2, 7) \times \mathbb{Z}_7$; 21 of the values of Table 5 rebuilt from their files (two searches, or a search from every vertex for the lifts and $(3, 11)$), with the quotients of the $(3, 13)$ and $(3, 14)$ lifts identified as the Desargues and Pappus graphs; $\alpha(S) = S$ for nine symmetric sets; and every number in Table 5 and Appendix B against its file. About three minutes.

The record values themselves were checked by the lab's coordinator with all-pairs breadth-first search (`research/coordinator_check/`); for the values of Section 7, see §7.7.

Appendix B. The values of Section 7, cell by cell

Tables 6–8 list the structures of Table 5: the number of elements, the 2013 table value where there is one, the group G (for the lifts: the base and the voltage group), the shape of S, $\mu = Ne^{-W/N}$ (§7.3; not given for the lifts) and the checks (§7.7). The shape $f + m \cdot k$ means f

elements fixed by α, e among them when $f \geq 1$, and m orbits of size k; "random" means S has no symmetry imposed; at $(11, 10)$ α is the field automorphism of §7.4. Each solution file is in `research/frontier/sols/` on `master` (Haar structures), `research/lifts/sols/` ($(4, 10)$, $(5, 10)$) or `research/synthetic/checks/haar/` (copied from the beyond branches); the edge lists of the lifts and of $(3, 11)$ are in `research/squads/own-small/records/`, `research/frontier/beyond/graphs/` and `research/synthetic/checks/haar/`. `check_haar.py` checks that every number here agrees with its file and generates these tables from one list.

(Δ, D)	elements	2013 table	G	S	μ	checks
$(5, 9)$	29,756	27,936	$\mathbb{Z}_{346} \rtimes_{29} \mathbb{Z}_{43}$	random	41.9	B, L, H
$(6, 9)$	127,296	117,360	$\mathrm{SL}(2, 17) \times \mathbb{Z}_{13}$	$0 + 2 \cdot 3$	29.7	B, L, H
$(7, 9)$	462,840	400,160	$\mathrm{PSL}(2, 29) \times \mathbb{Z}_{19}$	$1 + 2 \cdot 3$	38.2	B, L, H
$(8, 9)$	1,341,600	1,091,232	$\mathrm{SL}(2, 25) \times \mathbb{Z}_{43}$	$2 + 2 \cdot 3$	29.7	B, H
$(9, 9)$	3,499,776	2,961,536	$\mathrm{PSL}(2, 7) \times \mathrm{SL}(2, 7) \times \mathbb{Z}_{31}$	$0 + 3 \cdot 3$	30.5	B, X
$(10, 9)$	8,418,120	7,057,400	$\mathrm{PSL}(2, 59) \times \mathbb{Z}_{41}$	$0 + 2 \cdot 5$	42.4	P, I
$(11, 9)$	17,323,152	15,200,448	$\mathrm{SL}(2, 43) \times \mathbb{Z}_{109}$	$2 + 3 \cdot 3$	23.3	S, T
$(12, 9)$	35,445,648	30,001,152	$\mathrm{SL}(2, 83) \times \mathbb{Z}_{31}$	$0 + 4 \cdot 3$	29.5	S, T
$(13, 9)$	67,742,688	50,990,610	$\mathrm{SL}(2, 103) \times \mathbb{Z}_{31}$	$1 + 4 \cdot 3$	32.8	S, X
$(14, 9)$	120,043,728	95,087,738	$\mathrm{SL}(2, 107) \times \mathbb{Z}_{49}$	$2 + 4 \cdot 3$	24.2	S, T
$(15, 9)$	208,483,200	168,016,334	$\mathrm{SL}(2, 5) \times \mathrm{SL}(2, 19) \times \mathbb{Z}_{127}$	$0 + 5 \cdot 3$	24.9	G, I
$(16, 9)$	348,207,600	288,939,118	$\mathrm{SL}(2, 101) \times \mathbb{Z}_{169}$	$1 + 5 \cdot 3$	24.6	S, T
$(17, 9)$	501,707,400	–	$\mathrm{PSL}(2, 101) \times \mathbb{Z}_{487}$	$2 + 5 \cdot 3$	2.94	P, I
$(18, 9)$	790,626,816	–	$\mathrm{SL}(2, 127) \times \mathbb{Z}_{193}$	$3 + 5 \cdot 3$	2.85	S, T
$(19, 9)$	1,218,394,320	–	$\mathrm{SL}(2, 131) \times \mathbb{Z}_{271}$	$1 + 6 \cdot 3$	2.93	S, T
$(20, 9)$	1,823,175,864	–	$\mathrm{PSL}(2, 163) \times \mathbb{Z}_{421}$	$2 + 6 \cdot 3$	2.62	P, G
$(3, 10)$	810	800	$5 + 5$ base, $\mathbb{Z}_3 \times \mathrm{He}_3$	lift	–	A, L, H
$(4, 10)$	12,416	11,748	$\mathbb{Z}_{97} \rtimes_{19} \mathbb{Z}_{64}$	random	53.4	A, L, H
$(5, 10)$	95,472	90,068	$\mathbb{Z}_{1989} \rtimes_{461} \mathbb{Z}_{24}$	random	31.5	A, L, H
$(6, 10)$	524,160	452,032	$\mathrm{SL}(2, 5) \times \mathrm{PSL}(2, 7) \times \mathbb{Z}_{13}$	$0 + 2 \cdot 3$	23.6	B, L, H
$(7, 10)$	2,298,240	1,987,380	$\mathrm{SL}(2, 7) \times \mathrm{PSL}(2, 19)$	$1 + 2 \cdot 3$	30.9	B, I
$(8, 10)$	8,043,840	6,927,210	$\mathrm{PSL}(2, 7) \times \mathrm{PSL}(2, 19) \times \mathbb{Z}_7$	$2 + 2 \cdot 3$	33.2	B, I
$(9, 10)$	24,099,600	20,017,260	$\mathrm{PSL}(2, 151) \times \mathbb{Z}_7$	$0 + 3 \cdot 3$	35.7	P, I
$(10, 10)$	64,294,560	50,331,156	$\mathrm{PSL}(2, 81) \times \mathbb{Z}_{121}$	$0 + 2 \cdot 5$	41.6	B, I
$(11, 10)$	157,835,304	130,592,354	$\mathrm{PSL}(2, 243) \times \mathbb{Z}_{11}$	$1 + 2 \cdot 5$	60.6	G, I
$(12, 10)$	332,797,296	300,383,050	$\mathrm{SL}(2, 157) \times \mathbb{Z}_{43}$	$0 + 4 \cdot 3$	28.3	S, T
$(13, 10)$	683,943,456	617,330,936	$\mathrm{SL}(2, 137) \times \mathbb{Z}_{133}$	$1 + 4 \cdot 3$	24.3	S, T
$(14, 10)$	1,350,537,864	1,213,477,190	$\mathrm{PSL}(2, 157) \times \mathbb{Z}_{349}$	$2 + 4 \cdot 3$	27.6	P, G
$(15, 10)$	2,566,531,440	2,300,326,510	$\mathrm{SL}(2, 109) \times \mathbb{Z}_{991}$	$0 + 3 \cdot 5$	37.9	S, T, G
$(16, 10)$	4,708,566,360	4,119,507,330	$\mathrm{PSL}(2, 179) \times \mathbb{Z}_{821}$	$1 + 3 \cdot 5$	59.4	P, G
$(17, 10)$	7,022,052,336	–	$\mathrm{PSL}(2, 313) \times \mathbb{Z}_{229}$	$2 + 5 \cdot 3$	3.01	P, G
$(18, 10)$	11,787,972,672	–	$\mathrm{PSL}(2, 223) \times \mathbb{Z}_{1063}$	$3 + 5 \cdot 3$	3.06	G, P
$(19, 10)$	19,233,722,496	–	$\mathrm{SL}(2, 607) \times \mathbb{Z}_{43}$	$1 + 6 \cdot 3$	3.15	G, S
$(20, 10)$	30,518,778,360	–	$\mathrm{PSL}(2, 421) \times \mathbb{Z}_{409}$	$2 + 6 \cdot 3$	3.09	G, P

TABLE 6. The values at $D = 9$ and 10.

(Δ, D)	elements	2013 table	G	S	μ	checks
$(3, 11)$	1,474	–	$\mathbb{Z}_{67} \rtimes_9 \mathbb{Z}_{11}$	exhaustive	45.8	A*, H
$(4, 11)$	30,334	–	$\mathbb{Z}_{523} \rtimes_9 \mathbb{Z}_{29}$	random	44.1	B, A*, H
$(5, 11)$	311,426	–	$\mathbb{Z}_{5023} \rtimes_{2886} \mathbb{Z}_{31}$	random	19.6	B, A*, H
$(6, 11)$	2,122,518	–	$\mathbb{Z}_{20809} \rtimes_{12972} \mathbb{Z}_{51}$	random	10.7	B, H
$(7, 11)$	11,012,976	–	$\mathrm{SL}(2, 53) \times \mathbb{Z}_{37}$	$1 + 2 \cdot 3$	10.4	G, S
$(8, 11)$	45,329,808	–	$\mathrm{SL}(2, 283)$	$2 + 2 \cdot 3$	11	G, S
$(9, 11)$	155,250,816	–	$\mathrm{SL}(2, 223) \times \mathbb{Z}_7$	$0 + 3 \cdot 3$	10.6	G, S
$(10, 11)$	460,691,568	–	$\mathrm{SL}(2, 613)$	$1 + 3 \cdot 3$	9.26	G, S
$(11, 11)$	1,241,299,248	–	$\mathrm{SL}(2, 853)$	$2 + 3 \cdot 3$	10.4	G, S
$(12, 11)$	2,976,066,240	–	$\mathrm{SL}(2, 239) \times \mathbb{Z}_{109}$	$0 + 4 \cdot 3$	7.01	G, S
$(13, 11)$	6,781,213,392	–	$\mathrm{SL}(2, 563) \times \mathbb{Z}_{19}$	$1 + 4 \cdot 3$	7.56	G, S
$(14, 11)$	14,381,398,080	–	$\mathrm{SL}(2, 1009) \times \mathbb{Z}_7$	$2 + 4 \cdot 3$	6.87	G, S
$(15, 11)$	28,655,908,512	–	$\mathrm{SL}(2, 617) \times \mathbb{Z}_{61}$	$0 + 5 \cdot 3$	5.18	G, S
$(16, 11)$	29,961,590,400	–	$\mathrm{PSL}(2, 449) \times \mathbb{Z}_{331}$	$1 + 5 \cdot 3$	0	G*, P*
$(3, 12)$	2,368	–	$4 + 4$ base, $\mathbb{Z}_{37} \rtimes_6 \mathbb{Z}_8$	lift	–	A, A, L, H
$(4, 12)$	74,066	–	$\mathbb{Z}_{1277} \rtimes_{964} \mathbb{Z}_{29}$	random	28.3	A, M*, H
$(5, 12)$	1,058,342	–	$\mathbb{Z}_{8969} \rtimes_{205} \mathbb{Z}_{59}$	random	13.6	B, H
$(6, 12)$	9,366,240	–	$\mathrm{PSL}(2, 79) \times \mathbb{Z}_{19}$	$0 + 2 \cdot 3$	10.2	G, P
$(7, 12)$	59,838,504	–	$\mathrm{PSL}(2, 67) \times \mathbb{Z}_{199}$	$1 + 2 \cdot 3$	14.3	G, P
$(8, 12)$	284,095,056	–	$\mathrm{PSL}(2, 167) \times \mathbb{Z}_{61}$	$2 + 2 \cdot 3$	12.6	G, P
$(9, 12)$	1,105,785,120	–	$\mathrm{PSL}(2, 241) \times \mathbb{Z}_{79}$	$0 + 3 \cdot 3$	10.7	G, P
$(10, 12)$	3,712,424,352	–	$\mathrm{PSL}(2, 337) \times \mathbb{Z}_{97}$	$1 + 3 \cdot 3$	10.2	G, P
$(11, 12)$	11,189,148,096	–	$\mathrm{PSL}(2, 223) \times \mathbb{Z}_{1009}$	$2 + 3 \cdot 3$	13.3	G, P
$(12, 12)$	29,996,171,040	–	$\mathrm{PSL}(2, 241) \times \mathbb{Z}_{2143}$	$0 + 4 \cdot 3$	12.3	G, P
$(13, 12)$	29,997,644,040	–	$\mathrm{PSL}(2, 181) \times \mathbb{Z}_{5059}$	$1 + 4 \cdot 3$	0	G, P
$(14, 12)$	29,996,171,040	–	$\mathrm{PSL}(2, 241) \times \mathbb{Z}_{2143}$	$2 + 4 \cdot 3$	0	G, P

TABLE 7. The values at $D = 11$ and 12.

(Δ, D)	elements	2013 table	G	S	μ	checks
$(3, 13)$	3,400	–	Desargues, $\mathbb{Z}_{170}$	lift	–	A*, A, H
$(4, 13)$	192,382	–	$\mathbb{Z}_{2237} \rtimes_{1984} \mathbb{Z}_{43}$	random	24.2	A, M*, H
$(5, 13)$	3,594,910	–	$(\mathbb{Z}_{11} \rtimes_4 \mathbb{Z}_5) \times (\mathbb{Z}_{2971} \rtimes_{1764} \mathbb{Z}_{11})$	random	7.08	M*, H
$(6, 13)$	41,734,392	–	$\mathrm{PSL}(2, 83) \times \mathbb{Z}_{73}$	$0 + 2 \cdot 3$	9.29	G*, P′
$(7, 13)$	312,227,760	–	$\mathrm{SL}(2, 229) \times \mathbb{Z}_{13}$	$1 + 2 \cdot 3$	8.45	G*, S′
$(8, 13)$	1,739,514,000	–	$\mathrm{SL}(2, 499) \times \mathbb{Z}_7$	$2 + 2 \cdot 3$	7.52	G*, S′
$(9, 13)$	7,924,036,032	–	$\mathrm{SL}(2, 593) \times \mathbb{Z}_{19}$	$0 + 3 \cdot 3$	9.75	G*, S′
$(10, 13)$	19,980,294,768	–	$\mathrm{PSL}(2, 503) \times \mathbb{Z}_{157}$	$1 + 3 \cdot 3$	0.000154	G*, P′
$(3, 14)$	5,706	–	Pappus, $\mathbb{Z}_{317}$	lift	–	A*, A, H
$(4, 14)$	497,042	–	$\mathbb{Z}_{2731} \rtimes_{2400} \mathbb{Z}_{91}$	random	16.5	A, M*, H
$(5, 14)$	12,795,166	–	$\mathbb{Z}_{148781} \rtimes_{85175} \mathbb{Z}_{43}$	random	5.39	M*, H
$(6, 14)$	187,990,656	–	$\mathrm{SL}(2, 97) \times \mathbb{Z}_{103}$	$0 + 2 \cdot 3$	8.37	G*, S′
$(7, 14)$	1,674,490,944	–	$\mathrm{PSL}(2, 223) \times \mathbb{Z}_{151}$	$1 + 2 \cdot 3$	6.21	G*, P′
$(8, 14)$	11,325,238,056	–	$\mathrm{PSL}(2, 397) \times \mathbb{Z}_{181}$	$2 + 2 \cdot 3$	12.1	G*, P′

TABLE 8. The values at $D = 13$ and 14.

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